

Higher-Order Asymptotics of Regression-Adjusted Estimators in Randomized Trials*

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Abstract

Regression adjustment is widely used to estimate average treatment effects in randomized trials. Simple regression (REG, ANCOVA) and interacted regression (IREG, ANCOVA II) have the same first-order variance under balanced assignment, so first-order theory cannot choose between them. In a fixed-dimensional superpopulation framework we derive a closed form for the difference in the $1/n^2$ terms of their mean-squared-error expansions under complete randomization. It depends on covariate shape, treatment-effect heterogeneity, residual heteroskedasticity and nonlinear misspecification, and can favor either estimator; the resulting asymptotic deficiency ranges from a few observations to more than a hundred in our scalar designs with skewed covariates, and under extreme skewness the finite-sample gaps are large but the expansion no longer predicts their magnitude. As a practical companion, we show by simulation that without treatment-effect heterogeneity, interacted regression with HC1 standard errors can undercover substantially under skewed, heteroskedastic designs, in a pattern consistent with high within-arm leverage, while simple regression does not at the 95% level in our designs; HC3 largely corrects the variance scale but not the most extreme leverage configurations. We also note that the conventional robust variance of the centered interacted implementation omits a superpopulation variance component that matters when treatment-effect heterogeneity is large.

1 Introduction

Randomized controlled trials (RCTs) are fundamental to how causal effects are ascertained across a number of fields. One valuable attribute of RCTs is that they enable unbiased estimation of average treatment effects using simple methods such as the difference in means, without reliance on parametric assumptions [Imbens and Rubin, 2015]. However, while the difference-in-means estimator is unbiased, this does not automatically make it the preferred approach; and in particular regression adjustments are often used to improve power. For example, recent guidance from the U.S. Food and Drug Administration encourages adjustment for baseline covariates that are prognostic for the outcome [U.S. Food and Drug Administration, 2023].

The study of regression adjustments in RCTs has a long tradition in statistics. Fisher [1935] used arguments rooted in parametric linear modeling to advocate for such regression

*Code, simulation scripts and the computational certificates used in the proofs: <https://github.com/kelleyan-pixel/higher-order-regression-adjustment-for-rct/tree/v1.0.4> (release v1.0.4).

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adjustments. Fisher’s analysis was later critiqued by [Freedman \[2008\]](#), who argued that reliance on linear models was misaligned with model-free guarantees sought in RCT inference. [Lin \[2013\]](#), however, responded to Freedman’s critique by showing that regression adjustments do not require parametric assumptions to be near-unbiased for the average treatment effect. Furthermore, [Lin \[2013\]](#) showed that regression adjustments with full treatment-control interaction weakly dominate both the difference in means estimator and Fisher’s simple regression adjustment in terms of their leading-order (root- n scale) errors. Lin’s weak dominance result has led such treatment-interacted regressions to be often presented as a safe starting point for use of regression adjustments in RCTs [[Li and Ding, 2017](#), [Negi and Wooldridge, 2021](#), [Wager, 2025](#)].

A gap in this narrative, however, is that the first-order advantage of treatment-interacted regression over simple regression is strict only when the randomization probability satisfies $\pi \neq 0.5$ (and treatment effects vary with the covariates); the dominance is weak at every π . In evenly randomized trials with $\pi = 0.5$, the first-order accuracy of both methods is equal for any data-generating distribution, i.e., first-order theory offers no guidance on which method to prefer. And, while $\pi = 0.5$ is of course a special case, this is an important special case in applications, and it is not a priori clear we should prefer interacted regression to simple regression here. For example, [Zhao and Ding \[2023\]](#) emphasize the finite-sample trade-off associated with the larger number of parameters in interacted regression, arguing that simple regression may be more stable with limited sample size—especially when there are multiple treatment levels.

To see what may be at stake in practice, consider an example modeled after a typical tech industry setting. Say a company runs an A/B test on a sample of its user base and adjusts for X_i , each user’s own engagement level from the prior period (e.g., number of clicks). Such engagement measures are right-skewed because a small number of “super-users” generate far more activity than typical users. Suppose also that these high-engagement users generate more volatile outcomes, so that outcome noise increases with X_i . Specifically, let the raw X_i be marginally $\text{LogNormal}(0, 0.8^2)$ and let

$$\sigma(X_i) = 1 + 2X_i, \quad Y_i(0) = 0.9X_i + \sigma(X_i)u_i, \quad Y_i(1) = Y_i(0) + 1 + 0.4X_i,$$

with $u_i \stackrel{\text{iid}}{\sim} N(0, 1)$. Thus, the treatment effect itself varies with prior engagement, while the variance of the outcome is substantially larger for the users in the right tail. Here the whitened covariate has skewness 3.69 and kurtosis 34.4, and the heterogeneity share R_τ^2 defined in [Section 2](#) is 0.0032. Our second-order formula gives an asymptotic deficiency of $\Delta = 15.6$ for IREG relative to REG: asymptotically, using IREG costs about sixteen users’ worth of precision. At $n = 500$ this predicts that IREG’s mean-squared error should exceed REG’s by 3.1%, and simulation gives 2.3% (Monte Carlo standard error 0.02%); at $n = 2,000$ the prediction is 0.78% and simulation gives 0.67%. The HC1 intervals of both estimators are close to nominal here at $n = 500$: IREG+HC1 covers the population ATE in 94.5% of replications and REG+HC1 in 95.0%.

Two features of this example recur throughout the paper. First, the effect is real but bounded: a deficiency of sixteen observations matters at $n = 500$ and is negligible at $n = 50,000$. Second, the size of the deficiency is governed by high-order moments of the covariate distribution, and those same moments govern how quickly the second-order approximation becomes accurate. If we instead take $X_i \sim \text{LogNormal}(0, 1.8^2)$ with $\sigma(X_i) = 1 + 1.5X_i$ and a treatment-effect slope of 0.3—a tail heavy enough that the whitened covariate has skewness 136—then the formula returns $\Delta \approx 3.9 \times 10^4$ and predicts an MSE ratio of 78.5 at $n = 500$, whereas the simulated ratio is 1.68. In that regime the expansion gets the ordering right and substantially overstates the numerical gap; [Appendix C](#) quantifies where the approximation

stops predicting the magnitude of the gap in a closely related scalar family, with $\sigma(X) = 1 + X$ and slope 0.4, which differs from both examples in its heteroskedasticity.

Motivated by this gap, we conduct a higher-order asymptotic analysis of both simple and interacted regression adjustments in RCTs to elucidate the relative merits of the two approaches when $\pi = 0.5$. We frame our findings in terms of asymptotic deficiency [Hodges and Lehmann, 1970]. When two estimators have matching first-order behavior but one dominates the other on the second-order scale then, asymptotically, switching to the better one gives an accuracy boost that is equivalent to a fixed boost in sample size, which is referred to as the deficiency. Our results show that there is no universal second-order ordering: the comparison depends on covariate shape, treatment-effect heterogeneity, residual heteroskedasticity and nonlinear misspecification, and it can favor either estimator. The combination that favors simple regression in our examples is skewness interacting with heteroskedasticity or misspecification, as in the multiplicative error model commonly encountered in tech settings; simple regression can outperform interacted regression even when treatment-effect heterogeneity is absent. In the special case of Gaussian covariates with homoskedastic, well-specified errors the second-order coefficient instead favors IREG, and in the fixed- p asymptotic framework it is linear in the number of covariates p . This qualitative pattern runs against the finite-sample parameter-count heuristic emphasized by Zhao and Ding [2023], though the magnitude involved is exactly $(p+1)R_r^2$ observations in that case, and fixed- p asymptotics cannot speak to regimes in which the parameter count grows with n , which is the regime that motivates their concern.

As a practical companion to this theory, we study confidence intervals from standard software, `lm_lin` for IREG and `lm_robust` for REG from the R package `estimatr`. In simulations with $\gamma = 0$, where treatment effects are constant and there is no heterogeneity to account for, IREG with HC1 standard errors can undercover substantially when a covariate is skewed and residual variance grows in its tail, while REG with HC1 stays close to nominal at the 95% level in the designs we examine. The pattern is consistent with a leverage mechanism, since the interacted specification fits each arm separately. HC3 substantially improves IREG’s coverage in most of these designs, but it remains below nominal under very heavy-tailed covariates and is unreliable when a treatment arm contains an observation with leverage one. In the null-heterogeneity design with the most skewed covariate, HC1’s failure is primarily an understatement of the variance scale, which HC3 largely removes, leaving smaller distortions in the tails. Separately, under superpopulation sampling the variance of centered IREG for the population average treatment effect contains a component that its conventional robust variance does not include; it matters when covariate-explained heterogeneity is large. In a diagnostic, its feasible plug-in estimate tracks the true component when heterogeneity is large but is noisy and miscalibrated when heterogeneity is small or absent, so coverage gains from it need not reflect the estimand-level component (Section 3.2 and Appendix E).

1.1 Related Work

Regression adjustment in randomized experiments. The literature on regression adjustment in randomized experiments spans two complementary frameworks. The design-based literature treats the potential outcomes and covariates as fixed in a finite population, with randomness arising from treatment assignment. This is the framework of Lin [2013], Li and Ding [2017], and Zhao and Ding [2023]. A complementary superpopulation framework treats the experimental units themselves as random draws and studies repeated sampling over both the units and treatment assignment. Yang and Tsiatis [2001], Negi and Wooldridge [2021], Ye et al. [2023] and Wager [2025] work in this setting. Our analysis follows the latter framework. This distinction is important because our goal is to characterize higher-order sampling behavior

of REG and IREG under IID sampling rather than finite-population randomization error; as Section 3 shows, it also determines which variance estimator is appropriate.

The relative performance of REG and IREG has been studied primarily through first-order asymptotic efficiency. Yang and Tsiatis [2001] compare ANCOVA with and without treatment interactions in a superpopulation pretest-posttest setting, and Lin [2013] shows that IREG weakly dominates REG and the difference-in-means estimator in leading-order asymptotic variance for general treatment probabilities, with equality between REG and IREG under balanced treatment assignment. Zhao and Ding [2023] study the finite-sample consequences of the larger interacted specification and emphasize the role of the additional parameters introduced by treatment-covariate interactions, particularly in settings with multiple treatment levels. Our results complement this perspective: in the balanced Gaussian, homoskedastic, well-specified case the second-order coefficient favors IREG and, with p fixed as $n \rightarrow \infty$, is linear in p , while outside this setting the sign of the second-order comparison depends on skewness, tail behavior, heteroskedasticity and misspecification.

Variance estimation. That inference for a population average treatment effect after interacted regression requires an additional variance component relative to the finite-population case is discussed by Imbens and Wooldridge [2009], Negi and Wooldridge [2021], Zhao and Ding [2021] and Ye et al. [2023]; the last of these gives a systematic account of ANCOVA versus ANHECOVA for clinical trials. Samii and Aronow [2012] establish the equivalence between design-based Neyman variance estimators and heteroskedasticity-robust regression variances, which is why HC2 is the default in `estimat`r. Young [2019] documents concentrated leverage in regressions from published experiments, more so when treatment is interacted with covariates, and finds that randomization tests reject less often than the authors’ conventional tests. Finite-sample behavior of heteroskedasticity-robust variance estimators is studied by Chesher and Jewitt [1987], MacKinnon and White [1985], Long and Ervin [2000] and Imbens and Kolesár [2016]; Lei and Ding [2021] recommend HC3-type corrections for regression adjustment with many covariates.

Higher-order asymptotic theory. First-order asymptotic approximations cannot distinguish estimators with the same limiting distribution, even when their finite-sample behavior differs materially. Higher-order asymptotic theory addresses this through Edgeworth expansions and the associated moment (Nagar-type) expansions; see, for example, Bhattacharya and Rao [1976], Hall [1992] and Ghosh [1994], Sargan [1974] for conditions under which formal moment expansions describe trimmed exact moments, and Rothenberg [1988] for heteroskedasticity-robust test statistics. Hodges and Lehmann [1970] introduced asymptotic deficiency as a measure of the additional sample size required by a less accurate estimator, while Rothenberg [1984] emphasized the role of higher-order approximations when competing estimators share the same limiting distribution. Related work includes higher-order comparisons of first-order equivalent GMM and generalized empirical likelihood estimators by Newey and Smith [2004].

Recent work also studies higher-order regression adjustment under a design-based framework with many covariates. Lei and Ding [2021] characterize the bias of Lin’s estimator when p grows with n and propose a bias-corrected estimator, and Chiang et al. [2025], in Neyman’s design-based framework with the number of covariates growing more slowly than n , derive the first- and second-order terms of the stochastic expansions of regression-adjusted estimators, analyze the higher-order behavior of existing inference procedures, and propose a cross-fitted estimator and a modified HC3 standard error. Related work studies debiased and exactly bias-corrected adjustment [Lu et al., 2025, Chang et al., 2024], many regressors under covariate-adaptive randomization [Jiang et al., 2025], and estimators motivated by higher-order influence functions

[Zhao et al., 2024]. Closest in spirit to our analysis is Tan [2014], who compares first-order-equivalent regression and likelihood (calibration) estimators at second order in sampling and missing-data problems with a fixed number of auxiliary variables; the likelihood estimators are second-order unbiased but the regression estimators are not, and no estimator studied always has the smallest second-order variance, even after bias correction. Our comparison differs in the pair of estimators (pooled versus interacted linear working models), in the target (an average treatment effect) and in the design (complete randomization). Our analysis instead studies the fixed-dimensional superpopulation problem and focuses on the second-order MSE comparison between REG and IREG under balanced treatment assignment.

Estimands. Our expansion concerns the population ATE under IID superpopulation sampling. Much of the design-based literature, including Lin [2013], Lei and Ding [2021] and Chiang et al. [2025], targets the sample ATE of a fixed finite population, for which first-order variances differ and higher-order constants generally differ as well. Our own comparison of complete randomization with Bernoulli assignment shows how sensitive the n^{-2} constant is to the probabilistic framework (Theorem 11 shifts it by $4G$), so the constants of Theorem 2 should not be carried over to finite-population analyses without recomputation. The superpopulation view also matters for inference: the covariate mean at which IREG is centered is itself random, which is the source of the variance component discussed in Section 3.2.

2 Second-Order Analysis

We focus on completely randomized trials (CRTs) with causal effects formalized using the potential outcomes model [Imbens and Rubin, 2015]. We assume the covariates $X_i \in \mathbb{R}^p$ and potential outcomes $Y_i(0), Y_i(1) \in \mathbb{R}$ are drawn IID from a distribution P , with average treatment effect $\tau = \mathbb{E}[Y_i(1) - Y_i(0)]$. Treatment is assigned randomly conditional on the fixed treated group size $n_1 = \sum_{i=1}^n W_i = \lfloor \pi n \rfloor$; treatment assignments are exchangeable but not independent. Because both regression estimators include an intercept, translating the covariates does not change the estimators, so we work without loss of generality with centered covariates, $\mathbb{E}[X] = 0$.

The analyst observes (X_i, Y_i, W_i) for $i = 1, \dots, n$, where $Y_i = Y_i(W_i)$. We consider the following three estimators of the ATE:

$$\begin{aligned}\hat{\tau}_{DM} &= \frac{1}{|\{i : W_i = 1\}|} \sum_{\{i : W_i = 1\}} Y_i - \frac{1}{|\{i : W_i = 0\}|} \sum_{\{i : W_i = 0\}} Y_i, \\ \hat{\tau}_{REG} &= \hat{\tau}, \text{ with } Y_i \sim \alpha + X_i\beta + W_i\tau, \\ \hat{\tau}_{IREG} &= \hat{\tau} + \frac{1}{n} \sum_{i=1}^n X_i\hat{\delta}, \text{ with } Y_i \sim \alpha + X_i\beta + W_i\tau + W_iX_i\delta,\end{aligned}\tag{1}$$

where, e.g., $Y_i \sim \alpha + X_i\beta + W_i\tau$ means we algorithmically run a regression of Y_i against X_i and W_i with an intercept (without making any claims on well-specification of such a regression). To describe their first-order behavior it is helpful to define the best linear approximation to the conditional response function of the potential outcomes,

$$\left\{ \alpha_{(w)}^*, \beta_{(w)}^* \right\} = \underset{\alpha_{(w)}, \beta_{(w)}}{\operatorname{argmin}} \left\{ \mathbb{E} \left[(Y_i(w) - X_i\beta_{(w)} - \alpha_{(w)})^2 \right] \right\}, \tag{2}$$

together with the oracle residuals and the scaled heterogeneity direction

$$\begin{aligned}\varepsilon_i(w) &= Y_i(w) - \alpha_{(w)}^* - X_i^\top \beta_{(w)}^*, \\ Z_i &= \Sigma^{-1/2} X_i, \quad \Sigma = \text{Var}[X_i], \quad \gamma = \Sigma^{1/2} (\beta_{(1)}^* - \beta_{(0)}^*),\end{aligned}\tag{3}$$

so that $\mathbb{E}[Z_i] = 0$, $\mathbb{E}[Z_i Z_i^\top] = I_p$, $\mathbb{E}[Z_i \varepsilon_i(w)] = 0$ and $\varepsilon_i = \varepsilon_i(W_i)$. All population quantities below are invariant to the choice of matrix square root: replacing $\Sigma^{1/2}$ by $\Sigma^{1/2} O$ for an orthogonal O maps $Z \mapsto O^\top Z$ and $\gamma \mapsto O^\top \gamma$, under which every scalar in Theorem 2 is unchanged. The first-order behavior of the three estimators is well known [Negi and Wooldridge, 2021].

Proposition 1. *Suppose that X_i and $Y_i(w)$ have finite second moments and that $\text{Var}[X_i]$ is invertible. Then all three estimators above are asymptotically normal, $\sqrt{n}(\hat{\tau} - \tau) \Rightarrow \mathcal{N}(0, V)$, with limiting variances*

$$\begin{aligned}V_{DM} &= V_{IREG} + \text{Var} \left[X_i \left(\sqrt{\frac{\pi}{1-\pi}} \beta_{(0)}^* + \sqrt{\frac{1-\pi}{\pi}} \beta_{(1)}^* \right) \right], \\ V_{REG} &= V_{IREG} + \frac{(1-2\pi)^2}{\pi(1-\pi)} \text{Var} \left[X_i (\beta_{(1)}^* - \beta_{(0)}^*) \right], \\ V_{IREG} &= \text{Var} \left[X_i (\beta_{(1)}^* - \beta_{(0)}^*) \right] + \sum_{w=0,1} \frac{1}{\mathbb{P}[W_i = w]} \mathbb{E} \left[\left(Y_i(w) - X_i \beta_{(w)}^* - \alpha_{(w)}^* \right)^2 \right].\end{aligned}\tag{4}$$

Thus $\hat{\tau}_{IREG}$ weakly dominates both $\hat{\tau}_{DM}$ and $\hat{\tau}_{REG}$ at every π ; its advantage over $\hat{\tau}_{REG}$ is strict for generic distributions P when $\pi \neq 0.5$, while at $\pi = 0.5$ the two regression estimators have the same asymptotic variance

$$V^* = \|\gamma\|^2 + 4\mathbb{E}[\varepsilon_i^2] = \|\gamma\|^2 + 2\mathbb{E}[\varepsilon_i(1)^2] + 2\mathbb{E}[\varepsilon_i(0)^2].\tag{5}$$

This is the setting we study. Because first-order theory is indifferent between the two methods, we turn to second-order asymptotics.

2.1 Deficiency and the expansion MSE

Suppose we compare two estimators $\hat{\tau}_A$ and $\hat{\tau}_B$ of a shared quantity τ whose mean-squared errors agree on the (first-order) $1/n$ but not on the (second-order) $1/n^2$ scale,

$$\text{MSE}(\hat{\tau}_A) = \frac{V}{n} + \frac{c_A}{n^2} + o\left(\frac{1}{n^2}\right), \quad \text{MSE}(\hat{\tau}_B) = \frac{V}{n} + \frac{c_B}{n^2} + o\left(\frac{1}{n^2}\right).\tag{6}$$

Hodges and Lehmann [1970] propose quantifying the difference in performance by the number of extra samples required to compensate for using the less accurate estimator: they solve for k_n such that $\hat{\tau}_B$ run on k_n samples matches the accuracy of $\hat{\tau}_A$ on n samples as closely as possible. When $\lim_{n \rightarrow \infty} (k_n - n) = \Delta \in \mathbb{R}$ exists, Δ is the asymptotic deficiency of $\hat{\tau}_B$ relative to $\hat{\tau}_A$. Under (6),

$$\Delta = (c_B - c_A) / V.\tag{7}$$

Both regression estimators are smooth functions of sample moments, and for covariate distributions with atoms the relevant Gram matrices are singular with positive probability, so their exact second moments need not exist. We therefore work throughout with the second moment of the stochastic expansion rather than with the exact MSE, and we are explicit about the distinction.

Definition 1 (Expansion MSE). Let $\mathcal{V}_i$ collect the sample-moment coordinates entering the coefficient map of an estimator, let $\mu = \mathbb{E}[\mathcal{V}_i]$, $\bar{\mathcal{V}} = n^{-1} \sum_i \mathcal{V}_i$, and suppose $\hat{\tau} = g(\bar{\mathcal{V}})$ for a function g that is three times continuously differentiable in a neighborhood of μ with $g(\mu) = \tau$. Write $\zeta^{(n)} = \sqrt{n}(\bar{\mathcal{V}} - \mu)$ and let

$$\tilde{\tau}_n = \tau + \frac{1}{\sqrt{n}} a_i \zeta_i^{(n)} + \frac{1}{2n} H_{ij} \zeta_i^{(n)} \zeta_j^{(n)} + \frac{1}{6n^{3/2}} A_{3,ijk} \zeta_i^{(n)} \zeta_j^{(n)} \zeta_k^{(n)} \quad (8)$$

be the third-order Taylor polynomial of g at μ , with a , H and A_3 the first three derivative tensors of g at μ and the Einstein summation convention in force. The expansion MSE is $\text{MSE}^{\text{exp}}(\hat{\tau}) = \mathbb{E}[(\tilde{\tau}_n - \tau)^2]$.

Definition 1 presumes that the $\mathcal{V}_i$ are IID, as they are under Bernoulli assignment. Under complete randomization the assignments are dependent; there the observations in the two arms form independent IID samples (Lemma 8 in Appendix A), each estimator is a smooth function of the two per-arm sample means, and MSE^{exp} is the second moment of the third-order Taylor polynomial of that function at the per-arm population means. MSE^{exp} is a finite sum of moments of sample means and is well defined whenever those moments exist; it is the object that the deficiency calculations of Rothenberg [1984] and Hall [1992] manipulate. It is not the exact MSE. Conditions under which formal expansions of this kind describe suitably trimmed exact moments are given by Sargan [1974]; we do not verify them here, and our Monte Carlo checks use the untrimmed empirical MSE, reporting the frequency of degenerate draws (Appendix B).

Assumption 1 (Sampling and moments). $(X_i, Y_i(0), Y_i(1))$ are IID from P , $\Sigma = \text{Var}[X_i] \succ 0$, p is fixed, $\pi = 1/2$ with $n_1 = n/2$ for even n (and $n_1 = \lfloor n/2 \rfloor$ otherwise), and

$$\mathbb{E}[\|X_i\|^{12}] + \sum_{w=0}^1 \mathbb{E}[|Y_i(w)|^{12}] < \infty.$$

Assumption 1 is a convenient sufficient condition rather than a sharp one: the coefficients of Theorem 2 involve moments of Z of order at most four and mixed moments such as $\mathbb{E}[Z\varepsilon^2]$, while MSE^{exp} additionally requires the sixth moments of the moment vector $\mathcal{V}_i$, whose entries are products of at most two coordinates of (X_i, Y_i) . Assumption 1 implies $\mathbb{E}\|\mathcal{V}_i\|^6 < \infty$ by the Cauchy–Schwarz inequality.

2.2 The second-order difference

Theorem 2. Let Assumption 1 hold, and define

$$K = \mathbb{E}[ZZ^\top \|Z\|^2], \quad S = \mathbb{E}[Z\|Z\|^2], \quad S_\gamma = \mathbb{E}[ZZ^\top (Z^\top \gamma)],$$

$$\mathcal{E} = \mathbb{E}[ZZ^\top \varepsilon], \quad \mathcal{E}_W = \mathbb{E}[ZZ^\top (W - \tfrac{1}{2})\varepsilon] = \tfrac{1}{4}(\mathbb{E}[ZZ^\top \varepsilon(1)] - \mathbb{E}[ZZ^\top \varepsilon(0)]).$$

Here W is an auxiliary Bernoulli(1/2) variable independent of $(Z, \varepsilon(0), \varepsilon(1))$, and $\varepsilon = \varepsilon(W)$. Then, with MSE^{exp} as in Definition 1 and under the balanced CRT,

$$\text{MSE}^{\text{exp}}(\hat{\tau}_{\text{REG}}) - \text{MSE}^{\text{exp}}(\hat{\tau}_{\text{REG}}) = \frac{D_p}{n^2} + O(n^{-3}),$$

where

$$\begin{aligned} D_p = & - (2p+3)\|\gamma\|^2 + \gamma^\top K \gamma - (\gamma^\top S)^2 - 3\|S_\gamma\|_F^2 \\ & - 16 \text{tr}(S_\gamma \mathcal{E}_W) - 8(\gamma^\top S) \text{tr}(\mathcal{E}_W) + 8\gamma^\top \mathcal{E}_W S + 8 \text{tr}(\mathcal{E}^2) + 8S^\top \mathbb{E}[Z\varepsilon^2]. \end{aligned}$$

Consequently the asymptotic deficiency of IREG relative to REG, computed from the expansion MSE, is $\Delta = D_p/V^*$ with V^* as in (5).

The proof is in Appendix A. Under complete randomization the two arms are independent IID samples (Lemma 8), which yields an exact expansion of the conditional expansion MSE at any treatment fraction (Lemma 9); the proof then combines a reduction from the completely randomized design to Bernoulli assignment (Theorem 11) with the Bernoulli calculation (Theorem 12), whose index algebra is certified symbolically by a script in the replication repository. The remainder is $O(n^{-3})$ because Lemma 9 gives the complete-randomization expansion with an $O(n^{-3})$ remainder and the reduction identifies its n^{-2} coefficient exactly; the Monte Carlo sequences of Appendix B need not display this rate at the sample sizes used.

The expression for D_p combines four blocks: a covariate shape block scaled by the heterogeneity direction γ ; terms crossing γ with residual asymmetry; residual heteroskedasticity; and nonlinear misspecification. None of these alone dictates the ordering. In particular, the deficiency need not favor REG even when there is no treatment-effect heterogeneity: when $\gamma = 0$ all terms involving γ vanish and

$$D_p = 8 \operatorname{tr}(\mathcal{E}^2) + 8S^\top \mathbb{E}[Z\varepsilon^2].$$

While $\operatorname{tr}(\mathcal{E}^2) = \|E\|_F^2 \geq 0$, the term $S^\top \mathbb{E}[Z\varepsilon^2]$ can have either sign, so Δ can be positive or negative in this case.

Remark 3 (Bias and variance components). Two of the terms in D_p are a squared-bias difference; that regression adjustment has a bias of order $1/n$ is classical [Freedman, 2008, Lin, 2013]. Let $\tilde{\tau}_n$ be the Taylor polynomial of Definition 1, expanded under complete randomization in the per-arm sample means of Lemma 8. Its linear term has mean zero; its quadratic term has mean $\frac{1}{2} \operatorname{tr}(H\Sigma_n)$, where H is the Hessian of the estimator's coefficient map and Σ_n the covariance of the per-arm means, which is exactly proportional to $1/n$; and its cubic term has mean $O(n^{-2})$, because third central moments of sample means are third cumulants divided by squared sample sizes. The expansion bias is therefore $\mathbb{E}[\tilde{\tau}_n - \tau] = \frac{1}{2} \operatorname{tr}(H\Sigma_n) + O(n^{-2})$, and evaluating this leading term for the two estimators (Lemma 15 in Appendix A) gives

$$\mathbb{E}[\tilde{\tau}_{n,\text{REG}} - \tau] = -\frac{4 \operatorname{tr}(\mathcal{E}_W) + \gamma^\top S}{n} + O(n^{-2}), \quad \mathbb{E}[\tilde{\tau}_{n,\text{IREG}} - \tau] = -\frac{4 \operatorname{tr}(\mathcal{E}_W)}{n} + O(n^{-2}),$$

so that $n^2\{\text{Bias}_{\text{IREG}}^2 - \text{Bias}_{\text{REG}}^2\} = -(\gamma^\top S)^2 - 8(\gamma^\top S) \operatorname{tr}(\mathcal{E}_W)$ to leading order, exactly the $-Q - 8B_P$ part of D_p in the notation of Appendix A; the remaining terms are variance contributions. These are statements about the expansion, not about the exact finite-sample bias; Appendix B compares them with simulated biases.

To interpret the scale of Δ we use the heterogeneity share

$$R_\tau^2 = \frac{\|\gamma\|^2}{\|\gamma\|^2 + 4\mathbb{E}[\varepsilon_i^2]} = \frac{\|\gamma\|^2}{V^*}.$$

Consider the balanced doubly robust pseudo-outcome built from the linear working models $\mu_w(x) = \alpha_{(w)}^* + x^\top \beta_{(w)}^*$,

$$\phi_i = 2W_i\{Y_i - \mu_1(X_i)\} - 2(1 - W_i)\{Y_i - \mu_0(X_i)\} + \mu_1(X_i) - \mu_0(X_i).$$

Under the canonical representation $\phi_i - \tau = \gamma^\top Z_i + \xi_i$ with $\xi_i = 2W_i\varepsilon_i(1) - 2(1 - W_i)\varepsilon_i(0)$, we have $\mathbb{E}[Z_i\xi_i] = 0$, $\operatorname{Var}[\xi_i] = 4\mathbb{E}[\varepsilon_i^2]$ and $\operatorname{Var}[\gamma^\top Z_i] = \|\gamma\|^2$, so R_τ^2 is the population R^2 from linearly predicting the treatment-effect pseudo-outcome from Z_i . As noted in Section 3.2, R_τ^2 is also the fraction of the population-estimand sampling variance of $\hat{\tau}_{\text{REG}}$ that the default interacted-regression standard error omits under superpopulation sampling.

Corollary 4. *In addition to Assumption 1, suppose the potential outcomes satisfy a well-specified linear model with homoskedastic errors, $\mathbb{E}[\varepsilon_i(w) \mid X_i] = 0$ and $\mathbb{E}[\varepsilon_i^2(w) \mid X_i] = \sigma_w^2$ for $w \in \{0, 1\}$. Then $E = \mathcal{E}_W = 0$ and $\mathbb{E}[Z\varepsilon^2] = 0$, so*

$$\Delta = R_\tau^2 \left[-(2p+3) + \frac{\gamma^\top K \gamma - (\gamma^\top S)^2 - 3\|S_\gamma\|_F^2}{\|\gamma\|^2} \right]. \quad (9)$$

In the Gaussian sub-case $K = (p+2)I_p$, $S = 0$ and $S_\gamma = 0$, so that $\Delta = -(p+1)R_\tau^2$.

In the Gaussian homoskedastic case, then, the second-order coefficient favors IREG and is linear in the number of covariates, where p is held fixed as $n \rightarrow \infty$. This runs against the direction of the parameter-count heuristic of Zhao and Ding [2023], but its magnitude, $(p+1)R_\tau^2$ observations, is small relative to any realistic n ; and because Theorem 2 holds p fixed, it says nothing about designs in which p grows with n , which is the regime in which parameter counts are the binding concern [Lei and Ding, 2021, Chiang et al., 2025]. The following special case makes the mechanism transparent.

Remark 5 (Independent symmetric coordinates). Suppose that, in addition to the conditions of Corollary 4, the whitened covariates have independent coordinates with symmetric marginals and γ loads on coordinate j only. Then $S = 0$ and $S_\gamma = 0$, while $\gamma^\top K \gamma = (\kappa_j + p - 1)\|\gamma\|^2$ with $\kappa_j = \mathbb{E}[Z_j^4]$, so

$$\Delta = R_\tau^2 (\kappa_j - p - 4).$$

Simple regression is preferred at second order exactly when the kurtosis of the covariate driving the heterogeneity exceeds $p+4$; the Gaussian case $\kappa_j = 3$ recovers $\Delta = -(p+1)R_\tau^2$.

Corollary 6. *In addition to Assumption 1, suppose the potential outcomes follow*

$$Y_i(w) = \alpha_{(w)}^* + X_i^\top \beta_{(w)}^* + \eta_i \|Z_i\|,$$

where $\mathbb{E}[\eta_i] = 0$, $\mathbb{E}[\eta_i^2] = \nu^2$ and η_i is independent of (Z_i, W_i) . Then $\mathbb{E}[\varepsilon_i(w) \mid Z_i] = 0$ and $\mathbb{E}[\varepsilon_i^2(w) \mid Z_i] = \nu^2 \|Z_i\|^2$, where $\|\cdot\|$ is the Euclidean norm, so that $\mathbb{E}[Z_i \varepsilon_i^2(w)] = \nu^2 S$ and $\mathbb{E}[\varepsilon_i^2] = \nu^2 p$, and

$$\Delta = R_\tau^2 \left[-(2p+3) + \frac{\gamma^\top K \gamma - (\gamma^\top S)^2 - 3\|S_\gamma\|_F^2}{\|\gamma\|^2} \right] + \frac{8\nu^2 S^\top S}{\|\gamma\|^2 + 4\nu^2 p}. \quad (10)$$

Interpretation and scaling. It is useful to consider how the terms in (10) behave. The treatment-effect signal relative to residual variance is small in many applications, so terms multiplied by R_τ^2 rarely dominate unless the moment factors are large. The bracketed component is of order p in the leading cases we have examined—it equals $-(p+1)$ under Gaussian covariates and $\kappa_j - p - 4$ in the setting of Remark 5, which is negative unless the kurtosis exceeds $p+4$ —and its numerator grows with the higher moments of the covariate distribution, so it is sensitive to departures from Gaussianity in either direction.

The remaining term is nonnegative, since $8\nu^2 \|S\|^2 / (\|\gamma\|^2 + 4\nu^2 p) \geq 0$, and for $\nu > 0$ it vanishes if and only if $S = \mathbb{E}[Z\|Z\|^2] = 0$. Right-skewed covariates with a long upper tail, such as measures of user activity, engagement or spending, typically have large $\|S\|$, and the term can then dominate the covariate-shape block. The scalar case makes this transparent: with $p = 1$, Corollary 6 gives

$$\Delta = R_\tau^2 \left\{ \mathbb{E}[Z_i^4] - 4\mathbb{E}[Z_i^3]^2 - 5 \right\} + \frac{8\nu^2 \mathbb{E}[Z_i^3]^2}{\gamma^2 + 4\nu^2}.$$

Using Pearson's inequality $\mathbb{E}[Z_i^4] \geq \mathbb{E}[Z_i^3]^2 + 1$ [see, e.g., Rohatgi and Székely, 1989] and $8\nu^2/(\gamma^2 + 4\nu^2) = 2(1 - R_\tau^2)$,

$$\Delta \geq -4R_\tau^2 + (2 - 5R_\tau^2)\mathbb{E}[Z_i^3]^2.$$

So when $R_\tau^2 < 2/5$ the deficiency is positive as soon as $\mathbb{E}[Z_i^3]^2 > 4R_\tau^2/(2 - 5R_\tau^2)$, and the lower bound grows linearly in $\mathbb{E}[Z_i^3]^2$. The converse caveat, documented in Appendix C, is that the same third and fourth moments that make Δ large also slow the convergence of the expansion: in the lognormal family we study, the regime in which Δ is large is also the regime in which $1 + \Delta/n$ materially overstates the finite-sample MSE ratio at moderate n .

Remark 7 (Nearly balanced designs). Exact balance is not needed for Theorem 2 to be informative. Under Assumption 1 (with Lemma 9), if $n_1/n = 1/2 + h/\sqrt{n}$ for fixed h , then by Proposition 1 the first-order variance gap between the two estimators is $-16h^2\|\gamma\|^2/n + O(n^{-3/2})$, so that

$$\text{MSE}^{\text{exp}}(\hat{\tau}_{\text{IREG}}) - \text{MSE}^{\text{exp}}(\hat{\tau}_{\text{REG}}) = \frac{D_p - 16h^2\|\gamma\|^2}{n^2} + O(n^{-5/2}).$$

Imbalance of order $\sqrt{n}$ units therefore shifts the comparison toward IREG by a known amount; the $4\|\gamma\|^2$ term relating the Bernoulli and CRT designs in Theorem 11 is the special case in which h^2 is replaced by its expectation $1/4$ under Bernoulli randomization.

Estimating D_p . Theorem 2 suggests estimating D_p and choosing between REG and IREG accordingly. We do not propose such a rule: D_p depends on third and fourth moments and cross-moments, which a plug-in estimate recovers noisily, and choosing the estimator from the same sample creates a pretest problem whose effect on the final estimator is not described by the theorem.

3 Feasible Inference

The second-order results concern point estimation. In practice both estimators are reported with confidence intervals, and the behavior of those intervals depends on the variance estimator as much as on the estimator. This section studies the standard implementation. Its findings are simulation evidence, subordinate to the theory of Section 2.

3.1 Standard implementation

We use the R package `estimatr` [Blair et al., 2024]. REG is `lm_robust(y ~ w + x)`. IREG is `lm_lin(y ~ w, covariates = ~ x)`, which centers the covariates at their sample means, interacts them with treatment, and reports $\hat{\tau}_{\text{IREG}}$ as its coefficient on w . For a regression with k columns, residuals $\hat{e}_i$ and leverages H_{ii} , HC1 scales the squared residuals in the sandwich by $n/(n - k)$, HC2 by $1/(1 - H_{ii})$ and HC3 by $1/(1 - H_{ii})^2$; intervals use t critical values with $n - k$ degrees of freedom. HC1 is Stata's default, and Kranz [2024] finds that 98.1% of 40,571 heteroskedasticity-robust Stata regressions in reproduction packages of leading economics journals use it; HC2 is the default of `lm_robust` and `lm_lin` without clustering, motivated by its equivalence with Neyman's variance estimator [Samii and Aronow, 2012], so our HC1 results correspond to requesting `se_type = "HC1"` explicitly; HC3 approximates the jackknife [MacKinnon and White, 1985] and is recommended for small and moderate samples by Long and Ervin [2000]. The downward bias of the unadjusted sandwich is governed by the leverages [Chesher and Jewitt, 1987]. Other leverage-aware variance estimators, wild-bootstrap procedures and

degrees-of-freedom corrections [Imbens and Kolesár, 2016, Kranz, 2024] are related but outside our scope; we study the HC variants in common use. Leverage matters more for IREG than for REG in the same data, because the interacted specification amounts to separate least-squares fits in the two arms, each on about $n/2$ units.

3.2 A superpopulation caveat

For the centered IREG implementation considered here, the conventional robust covariance estimator treats $\bar{X}$ as fixed. Under the superpopulation sampling of Section 2, $\hat{\tau}_{\text{IREG}} = \hat{\tau} + \bar{X}^\top \hat{\delta}$ with $\bar{X}$ random, and the first-order sampling variance of $\hat{\tau}_{\text{IREG}}$ around the population ATE contains an additional component G/n that this estimator does not include, where

$$G = \left(\beta_{(1)}^* - \beta_{(0)}^* \right)^\top \Sigma_X \left(\beta_{(1)}^* - \beta_{(0)}^* \right) = \|\gamma\|^2$$

in the notation of (3), with $\Sigma_X = \Sigma = \text{Var}[X_i]$; equivalently $G = R_\tau^2 V^*$. The need for such a component is discussed by Imbens and Wooldridge [2009], Negi and Wooldridge [2021], Zhao and Ding [2021] and Ye et al. [2023]. The omission is specific to this implementation. To first order both estimators satisfy $\hat{\tau} - \tau = \gamma^\top \bar{Z} + \bar{\varepsilon}_1 - \bar{\varepsilon}_0 + o_P(n^{-1/2})$, with $\bar{Z}$ the full-sample mean of Z_i and $\bar{\varepsilon}_w$ the arm- w mean of ε_i , so both have first-order variance V^*/n . The residuals of the interacted fit are approximately ε_i , so its robust variance estimates $(V^* - G)/n$. REG omits the interactions, its residuals are approximately $\varepsilon_i + (W_i - \frac{1}{2})\gamma^\top Z_i$, and at $\pi = \frac{1}{2}$ its robust variance of the treatment coefficient is approximately $4n^{-2} \sum_i \hat{\varepsilon}_i^2$, which estimates $(4\mathbb{E}[\varepsilon_i^2] + G)/n = V^*/n$; in the large-heterogeneity design of Appendix E, REG+HC1 covers the population ATE in 95.2% of replications. Omitting G/n gives a nominal 95% IREG interval asymptotic coverage $2\Phi(1.96\sqrt{1 - R_\tau^2}) - 1$, which is 94.9% at $R_\tau^2 = 0.01$ and 93.7% at $R_\tau^2 = 0.1$: the component matters when covariate-explained heterogeneity is large relative to outcome noise and is negligible when it is small. A feasible first-order correction replaces G by its plug-in estimate,

$$\hat{V}_{\text{SP}} = \hat{V}_{\text{IREG}} + \frac{\hat{\delta}^\top \hat{\Sigma}_X \hat{\delta}}{n}, \quad (11)$$

with $\hat{\delta}$ the fitted interaction coefficients and $\hat{\Sigma}_X$ the sample covariance of X .

The plug-in must be distinguished from the quantity it estimates, because it is biased upward in finite samples. To leading order $\mathbb{E}[\hat{\delta}^\top \hat{\Sigma}_X \hat{\delta}] \approx G + \text{tr}\{\Sigma_X \text{Var}[\hat{\delta}]\}$. With homoskedastic, well-specified errors of variance σ^2 in both arms, each arm's slope is estimated from about $n/2$ units, so $\text{Var}[\hat{\delta}] \approx 4\sigma^2 \Sigma_X^{-1}/n$ and

$$\mathbb{E}[\hat{\delta}^\top \hat{\Sigma}_X \hat{\delta}/n] \approx \frac{G}{n} + \frac{4p\sigma^2}{n^2}.$$

In the five Gaussian homoskedastic designs of the correction study (Appendix E) this approximation matches the simulated mean of the plug-in to within 0.8%. The second term does not depend on G : when $\gamma = 0$ the plug-in consists of it entirely, and when G is small it can exceed G/n . Under heteroskedasticity and heavy-tailed covariates $\text{Var}[\hat{\delta}]$ is larger, its leading form involving $\mathbb{E}[\varepsilon_i^2(X_i - \mu)(X_i - \mu)^\top]$, and the leading-order approximation itself becomes less accurate; we use it as an explanation of the mechanism, not as an exact formula. A coverage change produced by (11) therefore need not reflect the estimand-level component.

Appendix E separates the two by also adding the true G/n (an oracle correction, not a feasible procedure) on the same simulated datasets, at nominal levels from 80% to 99%. With

large heterogeneity ($R_\tau^2 = 0.692$) the plug-in averages 1.00 times G/n , and the feasible and oracle intervals are both close to nominal at every level examined: at 95%, HC1 covers 72.7%, HC1 plus G/n 95.1% and HC1 plus the plug-in 95.2%, with a median interval 1.80 times as wide as HC1's. With $\gamma = 0$ and a heavily skewed covariate, adding $G/n = 0$ changes nothing, while the plug-in, which widens the median interval by a factor of only 1.07, moves HC1 coverage at 95% from 83.0% to 95.1%. That agreement with nominal is a crossing point, not calibration: across nominal levels the plug-in interval is miscalibrated, covering 74.6% at 80% and 99.5% at 99%. Since $G = 0$ in this design, the plug-in's effect is finite-sample compensation for the understated HC1 variance, not recovery of a missing variance component. We therefore use the standard implementation in the rest of this section and do not propose (11) as a general remedy.

3.3 Coverage without treatment-effect heterogeneity

The finite-sample phenomenon we study is distinct from the caveat above. Every design in this section has $\gamma = 0$: $\beta_{(1)}^* = \beta_{(0)}^*$ and the two potential outcomes share their residual, so every unit has the same treatment effect $\tau = 1$. Then $R_\tau^2 = 0$ and $G = 0$ exactly, the sample and population average treatment effects coincide, and the component of Section 3.2 is identically zero. Undercoverage in these designs is therefore a finite-sample failure of the variance estimator, not an omitted estimand-level variance.

Designs. Here u_i denotes simulation noise. Treatment is assigned by a balanced CRT with $n = 500$ unless stated, and

$$Y_i(0) = 0.9 \sum_j X_{ij} + g(X_i) + \sigma(X_i)u_i, \quad Y_i(1) = Y_i(0) + 1, \quad u_i \sim N(0, 1).$$

The base design has $p = 1$, a raw right-skewed covariate $X_i \sim \text{LogNormal}(0, s^2)$ such as prior engagement, $\sigma(X) = 1 + X$ and $g = 0$. We vary s , add misspecification g with homoskedastic or heteroskedastic noise, vary n , and use $p = 5$ with X_{i1} lognormal, $X_{i2}, \dots, X_{i5} \sim N(0, 1)$ and $\sigma = 1 + X_{i1}$. Each design uses 20,000 replications (10,000 at $n = 2,000$ and 4,000 at $n = 8,000$); the Monte Carlo standard error of every reported coverage is at most 0.27 percentage points with 20,000 replications and at most 0.36 in the rows with fewer replications. The simulations call `estimatr`'s own fitting routine with design matrices built exactly as `lm_robust` and `lm_lin` build them; Appendix D describes how this was validated.

Results. Table 1 shows that IREG+HC1 coverage deteriorates steadily as the covariate becomes more skewed: from 94.5% at $s = 0.6$ to 89.9% at $s = 1.5$ and 83.0% at $s = 2.0$. At the 95% level, REG+HC1 stays between 94.7% and 95.4% in every design of this section. HC2 recovers part of the gap for IREG and HC3 most of it. Table 2 extends the picture. With Gaussian covariates, quadratic misspecification is harmless; with a skewed covariate, a misspecification that grows in the tail produces undercoverage even with homoskedastic noise (93.0% for IREG+HC1). In the base design with $s = 1.2$, the undercoverage shrinks as n grows, from 91.3% at $n = 200$ to 94.6% at $n = 2,000$ and 94.5% at $n = 8,000$, and it persists with $p = 5$.

The pattern is consistent with the leverage mechanism of Chesher and Jewitt [1987]. IREG is equivalent to separate least-squares fits in the two arms, each on $n/2$ units, so a rare extreme covariate value has roughly twice the leverage it would have in REG's pooled fit—the median of the largest IREG leverage rises from 0.15 at $s = 0.6$ to 0.65 at $s = 2.0$ —and its arm's slope is pulled towards it, shrinking its residual exactly where $\sigma(X)$ is largest. HC1 does not undo this shrinkage; the factor $1/(1 - H_{ii})^2$ in HC3 largely does.

Scale or shape. Table 3 examines the $s = 2$ design more closely. If the studentized statistic $T = (\hat{\tau} - \tau)/\widehat{\text{SE}}$ behaved like a scaled t variable, $T \approx c t_{n-k}$, a single scale c read off the 95% coverage would predict coverage at every level. For IREG+HC1, $c = 1.43$, and the prediction is within 1.4 points of the observed coverage at every level from 80% to 99%: HC1’s failure in this design is primarily an understatement of the variance scale. The ratio of the empirical $|T|$ quantile to the t quantile nevertheless rises from 1.43 at 80% to 1.74 at 99%, so the tails of T are also heavier than those of a scaled t . REG+HC1 shows a milder form of the same crossing pattern in this design: it covers 76.3% at 80% and 99.5% at 99%, with the $|T|$ quantile ratio falling from 1.08 to 0.94, so its near-nominal coverage at 95% is also partly a crossing point; at 80% IREG+HC3 (79.5%) is closer to nominal.

Variability or standard error? Coverage alone does not say whether IREG is more variable or only has a poorer standard error; Table 4 reports both. At $s = 2$, the Monte Carlo standard deviation of $\hat{\tau}_{\text{IREG}}$ is 3.48 against 2.46 for $\hat{\tau}_{\text{REG}}$, an MSE ratio of 2.00, while their median HC1 standard errors are almost equal (1.75 and 1.73): in this design IREG is genuinely more variable, and HC1 does not register the difference. Because both sampling distributions are heavy-tailed, the Monte Carlo standard deviations exceed the median standard errors even for REG, whose coverage is close to nominal, so the informative comparison is between the two estimators.

Connection with Section 2. The same covariate-tail features that enter the second-order MSE coefficient are also associated with more severe within-arm leverage and HC1 undercoverage in our simulations. In the designs of this section $\gamma = 0$, so every term of D_p that involves γ vanishes, including the kurtosis terms; with one covariate and well-specified, shared residuals, only the product of the covariate’s skewness and the heteroskedasticity moment $\mathbb{E}[Z\varepsilon^2]$ remains, and it is positive, favoring REG. Table 4 compares the implied prediction $1 + \Delta/n$ with the simulated MSE ratio in the same designs: they agree closely for $s \leq 0.8$ (1.03 predicted against 1.02 simulated at $s = 0.8$), the simulated ratio exceeds one in every design, as predicted, and from $s = 1$ onward the prediction exceeds the simulated ratio by a growing margin, for the reasons discussed in Appendix C. The covariate representation matters accordingly. On the same $s = 2$ datasets, adjusting for $\log(1 + X)$ instead of X lowers the median of the largest IREG leverage from 0.65 to 0.08, and IREG+HC1 then covers 95.6% instead of 83.0%; but the median interval widens from 6.87 to 9.92, because the outcome is linear in X in this design. The check ties the extreme undercoverage to the tail of the raw covariate; it does not show that transformation is generally preferable.

Table 1: Coverage (%) of nominal 95% intervals when $\gamma = 0$ ($R_\tau^2 = 0$): $X \sim \text{LogNormal}(0, s^2)$, $\sigma(X) = 1 + X$, $p = 1$, $n = 500$, 20,000 replications. “Max. lev.” is the median over replications of the largest IREG leverage.

s	skew Z	Max. lev.	IREG (lm_lin)			REG (lm_robust)	
			HC1	HC2	HC3	HC1	HC3
0.6	2.3	0.15	94.5	94.7	94.8	94.7	94.9
0.8	3.7	0.21	94.4	94.6	95.1	94.7	95.1
1.0	6.2	0.29	94.2	94.7	95.4	95.1	95.8
1.2	11.2	0.37	92.6	93.7	95.1	94.8	95.8
1.5	33.5	0.48	89.9	91.9	94.7	95.0	96.7
1.8	136.4	0.59	86.1	89.2	94.1	95.1	97.4
2.0	414.4	0.65	83.0	87.1	93.9	95.3	97.9

Table 2: Coverage (%) of nominal 95% intervals when $\gamma = 0$ in further designs. Unless stated, $p = 1$ and $X \sim \text{LogNormal}(0, s^2)$; Z is the standardized covariate.

Design	n	Reps	IREG			REG		
			HC1	HC2	HC3	HC1	HC3	
<i>Misspecification, homoskedastic unless noted</i>								
Gaussian X , $0.3(X^2 - 1)$	500	20,000	94.9	94.9	95.0	94.9	95.0	
$s = 1$, $2 \log X$	500	20,000	94.5	95.0	95.7	95.0	95.6	
$s = 1$, $0.3(Z^2 - 1)$	500	20,000	93.0	94.5	95.7	95.1	96.2	
$s = 1$, $2 \log X$, $\sigma = 1 + X$	500	20,000	94.2	94.7	95.5	94.9	95.6	
<i>Sample size</i>								
$s = 1.2$, $\sigma = 1 + X$	200	20,000	91.3	92.8	94.9	94.8	96.4	
$s = 1.2$, $\sigma = 1 + X$	500	20,000	92.6	93.7	95.1	94.8	95.8	
$s = 1.2$, $\sigma = 1 + X$	2,000	10,000	94.6	94.9	95.6	95.4	95.8	
$s = 1.2$, $\sigma = 1 + X$	8,000	4,000	94.5	94.8	94.9	94.8	95.0	
<i>More covariates</i>								
$p = 5$, $s = 1$, $\sigma = 1 + X_1$	500	20,000	94.2	94.7	95.5	95.1	95.7	
$p = 5$, $s = 1$, $\sigma = 1 + X_1$	200	20,000	93.1	93.8	95.5	94.9	96.1	

3.4 HC3 and extreme leverage

HC3 largely removes the scale error. In the $s = 2$ design its implied scale is 1.05 (Table 3), and the $|T|$ quantile ratio is 1.01 at 80%, rising to 1.14 at 99%. The remaining deviations are consistent with higher-order distortion of the studentized statistic beyond a simple scale error, which a different leverage adjustment of the variance need not remove. Table 5 shows what the adjustment costs. Along the skewness sweep, the median ratio of the HC3 to the HC1 standard error of $\hat{\tau}_{\text{IREG}}$ rises from 1.01 at $s = 0.6$ to 1.28 at $s = 2$, with a 90th percentile of 1.92; for REG the corresponding values at $s = 2$ are 1.09 and 1.34. The median HC1 widths of IREG and REG are nearly equal throughout (6.87 and 6.80 at $s = 2$), so IREG+HC1’s undercoverage is not accompanied by narrower typical intervals, and the gap between the median and the 90th percentile of the HC3/HC1 ratio shows that HC3’s adjustment is concentrated in a minority of replications. HC3 remains below nominal under very heavy-tailed covariates (Table 1, $s \geq 1.8$).

Leverage one. An observation with an extreme covariate value can carry a large share of the variance of $\hat{\tau}_{\text{IREG}}$. When a treatment arm contains exactly one such observation, it has leverage one in that arm’s fit and is fitted exactly, so its residual is zero. A residual-based variance estimator then has no information about the observation’s contribution: HC1 counts it as zero, and HC2 and HC3 face a 0/0 that no leverage adjustment can resolve. To isolate this we use a rare-value design: $p = 1$, $n = 500$, $X_{i1} = \sqrt{(1 - \rho)/\rho}$ with probability $\rho = 0.02$ and $-\sqrt{\rho/(1 - \rho)}$ otherwise, and $\sigma = 1 + |X_{i1}|$ or $\sigma = 1$. An arm contains exactly one rare unit, and the other arm at least one, in 6.1% of replications (exact probability 6.39%) (Table 6; each leverage-one subset contains at least 1225 replications, so the Monte Carlo standard errors of its coverages are at most 1.4 points). With $\sigma = 1 + |X_{i1}|$ the isolated unit also carries the largest residual variance, and IREG+HC1 covers 85.6% of those replications; with $\sigma = 1$, IREG+HC1 covers 95.7% of those replications. In 1.4% of replications (exact probability 1.28%) an arm contains no rare unit and the interacted design is rank-deficient; these replications are included in the “all replications” rows as the software reports them. REG estimates a single slope from

Table 3: Scale versus shape of the studentized statistic for IREG and REG in the $\gamma = 0$, $s = 2$ design of Table 1. The implied scale c solves $2F_t(q_{0.975}/c) - 1 = \text{observed 95\% coverage}$; “predicted” is $2F_t(q/c) - 1$ at each level; the last row of each block divides the empirical quantile of $|T|$ by the corresponding t quantile.

	80%	90%	95%	98%	99%
<i>IREG, HC1</i> (implied scale $c = 1.43$)					
observed coverage	64.4	75.5	83.0	89.1	91.9
predicted by scale c	63.0	75.0	83.0	89.7	92.9
$ T $ quantile / t quantile	1.43	1.46	1.52	1.64	1.74
<i>IREG, HC3</i> (implied scale $c = 1.05$)					
observed coverage	79.5	89.2	93.9	96.9	98.0
predicted by scale c	77.9	88.4	93.9	97.4	98.6
$ T $ quantile / t quantile	1.01	1.02	1.05	1.10	1.14
<i>REG, HC1</i> (implied scale $c = 0.99$)					
observed coverage	76.3	88.8	95.3	98.6	99.5
predicted by scale c	80.6	90.5	95.3	98.2	99.1
$ T $ quantile / t quantile	1.08	1.03	0.99	0.95	0.94

Table 4: Monte Carlo standard deviation and MSE of $\hat{\tau}_{\text{IREG}}$ and $\hat{\tau}_{\text{REG}}$ against their median HC1 and HC3 standard errors, in the designs of Table 1; $\sqrt{V^*/n}$ is the common first-order standard deviation of the two estimators and $1 + \Delta/n$ the second-order prediction of Theorem 2; Monte Carlo standard errors of the simulated MSE ratio are in parentheses, in units of the last digit.

s	Standard deviation			MSE ratio IREG/REG		Median standard error			
	IREG	REG	$\sqrt{V^*/n}$	simulated	$1 + \Delta/n$	IREG HC1	REG HC1	IREG HC3	REG HC3
0.6	0.211	0.210	0.209	1.007 (<1)	1.01	0.207	0.207	0.210	
0.8	0.242	0.240	0.243	1.017 (1)	1.03	0.237	0.237	0.243	
1.0	0.301	0.295	0.306	1.043 (2)	1.09	0.288	0.287	0.301	
1.2	0.420	0.401	0.428	1.096 (3)	1.34	0.373	0.371	0.402	
1.5	0.812	0.721	0.882	1.271 (14)	4.81	0.618	0.614	0.714	
1.8	1.864	1.464	2.303	1.620 (39)	72	1.128	1.118	1.414	
2.0	3.482	2.465	4.896	1.996 (56)	672	1.748	1.731	2.327	

Table 5: Median width of nominal 95% intervals and the ratio $\text{SE}_{\text{HC3}}(\hat{\tau})/\text{SE}_{\text{HC1}}(\hat{\tau})$ (median and 90th percentile over replications) in the designs of Table 1.

s	Median width				IREG HC3/HC1		REG HC3/HC1	
	IREG HC1	IREG HC3	REG HC1	REG HC3	median	p90	median	p90
0.6	0.82	0.82	0.81	0.82	1.01	1.02	1.01	1.01
0.8	0.93	0.95	0.93	0.94	1.02	1.05	1.01	1.03
1.0	1.13	1.18	1.13	1.15	1.04	1.11	1.02	1.05
1.2	1.47	1.58	1.46	1.51	1.07	1.22	1.03	1.10
1.5	2.43	2.81	2.41	2.57	1.13	1.45	1.05	1.19
1.8	4.43	5.56	4.39	4.81	1.22	1.71	1.07	1.28
2.0	6.87	9.14	6.80	7.57	1.28	1.92	1.09	1.34

both arms and does not create the same arm-specific leverage problem induced by the interacted specification; it stays near nominal at 95% in every row of the table.

Software. What HC2 and HC3 report at leverage one depends on how software evaluates the 0/0, and packages do not treat full-leverage observations consistently [Kranz, 2024]. Table 6 reports two versions of `estimatr`, which handle the case differently (Appendix D); neither recovers the variance contribution of the isolated observation.

Table 6: Coverage (%) in the rare-value design ($\gamma = 0$, $p = 1$, $n = 500$, $\rho = 0.02$, 20,000 replications). Rows: all replications as the software reports them; replications with a full-rank interacted design; and, within those, replications in which an arm contains a leverage-one unit. With $\sigma = 1 + |X_1|$ these are 1.4% rank-deficient and 6.1% leverage-one replications; with $\sigma = 1$, 1.3% and 6.6%; the exact probabilities are 1.28% and 6.39%. The two `estimatr` versions are applied to the same simulated datasets; they differ only in how they evaluate the 0/0 HC2/HC3 contribution and in which collinear column they drop in rank-deficient fits. REG does not depend on the version. A replication without a finite interval counts as a non-cover.

	IREG, v1.0.2			IREG, v2.0.0			REG	
	HC1	HC2	HC3	HC1	HC2	HC3	HC1	HC3
<i>Rare value, $\sigma = 1 + X_1$</i>								
all replications	92.0	90.1	94.8	92.0	93.1	94.6	94.9	96.0
full-rank designs	92.2	90.5	95.1	92.2	93.3	94.9	94.9	96.0
of which leverage-one	85.6	40.2	94.9	85.6	85.8	90.5	95.7	96.6
<i>Rare value, $\sigma = 1$</i>								
all replications	94.7	90.9	94.7	94.7	94.7	94.9	95.0	95.0
full-rank designs	94.9	91.2	95.2	94.9	94.9	95.1	94.9	95.0
of which leverage-one	95.7	41.3	97.6	95.7	95.7	97.1	95.8	95.9

3.5 Practical synthesis

Across the designs considered, the point-estimation and inference results suggest two practically distinct regimes. When treatment-effect heterogeneity is small, the second-order comparison is driven by covariate shape, heteroskedasticity and misspecification, and in several of our skewed, heteroskedastic designs it favors REG; IREG can meanwhile encounter finite-sample leverage problems that make robust inference difficult, although HC3 substantially improves coverage in most of these designs. In this regime G/n is small or zero, and the plug-in’s bias term can be large relative to it. When heterogeneity is large, the second-order comparison moves in IREG’s favor, and G/n becomes an important part of the uncertainty about the population ATE for the centered IREG implementation. In our large-heterogeneity design, which is Gaussian, homoskedastic and well specified, Corollary 4 gives a deficiency of 4.2 observations in IREG’s favor, a real but modest second-order gain; the feasible correction closely tracks the oracle and gives near-nominal coverage at every nominal level examined, with intervals a median 1.80 times as wide as HC1’s; and REG+HC1 covers the population ATE in 95.2% of replications without any correction. These patterns are a synthesis of the designs studied rather than a decision rule: the second-order comparison also depends on covariate moments, heteroskedasticity and misspecification, and R_τ^2 is not directly observed.

In our designs the undercoverage of IREG+HC1 grows steadily with the covariate tail, well before any observation reaches leverage one, together with the largest within-arm leverage and

the ratio of the HC3 to the HC1 standard error (Tables 1 and 5). Comparing the two standard errors of $\hat{\tau}_{\text{IREG}}$ is therefore a simple descriptive check of how sensitive the reported uncertainty is to leverage; we do not propose a threshold. Comparisons between the two estimators use the same variance estimator (Tables 1, 4 and 5); pairings such as IREG+HC3 against REG+HC1 mix the choice of adjustment with the choice of variance estimator and are not head-to-head comparisons. None of this is a general ranking of the two estimators.

4 Discussion

Regression adjustment in balanced trials involves three mechanisms that are easy to conflate, and the paper speaks to each separately.

Point-estimator efficiency. REG and IREG have the same first-order variance when $\pi = 1/2$, and Theorem 2 gives the next term of the comparison. It depends on treatment-effect heterogeneity, covariate shape, residual heteroskedasticity and nonlinear misspecification, and it can favor either estimator. Under Gaussian covariates with homoskedastic, well-specified errors the second-order coefficient favors IREG and is linear in p (for fixed p); with independent symmetric coordinates the ordering is decided by whether the kurtosis of the heterogeneity-driving covariate exceeds $p + 4$; heteroskedasticity or misspecification add terms that can favor REG even without heterogeneity. The deficiency can be practically substantial: in the scalar lognormal family of Table 9 it is 12 observations at $s = 0.8$, 41 at $s = 1$ and 166 at $s = 1.2$, and at $n = 500$ the simulated MSE of IREG exceeds that of REG by 4.3% and 9.2% in the last two designs. At $n = 500$ the expansion overstates these gaps (it predicts 8.3% and 33.2%), but the agreement improves with n : at $n = 8,000$ it predicts 0.5% against a simulated 0.4% for $s = 1$. In the most heavy-tailed designs the finite-sample MSE gaps are larger still (the MSE ratio is 1.68 (Monte Carlo SE 0.033) in the heavy-tailed example of Section 1 and 2.00 (SE 0.056) in the $\gamma = 0$, $s = 2$ design of Section 3.3), but the n^{-2} expansion no longer predicts their magnitude (Appendix C). The sign of the comparison agrees with the expansion in every design where we compare the two (Tables 4, 7 and 9 and the heavy-tailed example), so the decomposition of D_p remains informative about direction beyond the range in which it is accurate about magnitude. Which estimator is preferable at second order thus depends on features of the data-generating process that first-order theory cannot see.

Estimand-level uncertainty. For the population ATE, the first-order variance of the centered IREG estimator includes $G/n = R_\tau^2 V^*/n$, which its conventional robust variance does not include; REG’s robust variance has no such omission. This concerns what is being estimated, not the quality of a variance estimator. The feasible plug-in (11) is biased upward by a term of leading order $\text{tr}\{\Sigma_X \text{Var}[\hat{\delta}]\}/n$ that does not depend on G . When G is small or zero this term can be large relative to the true component, so the plug-in can change coverage substantially for reasons unrelated to the estimand; a small G does not make the feasible correction harmless.

Finite-sample variance estimation. Even with $G = 0$, the robust variance of IREG can be too small in finite samples. The pattern in our simulations is consistent with the leverage mechanism of Chesher and Jewitt [1987]: the interacted specification creates high within-arm leverage for rare covariate values, and HC1 does not adjust for leverage. HC3 substantially improves coverage in most of our designs, but not under very heavy-tailed covariates, and at leverage-one observations HC2 and HC3 face a 0/0 that software resolves differently. REG does not create the same arm-specific leverage problem, and its HC1 intervals are close to nominal at the 95% level in the designs we examine, although in the most heavy-tailed designs they too are miscalibrated across nominal levels.

Section 3.5 summarizes how these mechanisms co-vary with the size of treatment-effect heterogeneity in the designs we study.

None of these mechanisms explains the others. For practice, the upshot is that asymptotic efficiency of the point estimator does not determine the quality of the resulting confidence interval: the variance estimator actually used, the leverage structure of the fitted design, and the intended estimand each matter.

Several limitations remain. Our theorem holds p fixed and does not cover settings in which the number of covariates grows with the sample size, the regime studied by Lei and Ding [2021] and Chiang et al. [2025] and the one in which parameter-count heuristics are most relevant. The

second-order result is a statement about the expansion MSE of Definition 1; relating it to exact or trimmed moments requires conditions we do not verify. The inference results are numerical: we do not derive higher-order expansions for HC coverage in the sense of Rothenberg [1988], and the leverage-one results depend on the software’s treatment of a 0/0 contribution. We treat the superpopulation variance component of IREG only briefly (Section 3.2 and Appendix E); the upward bias of the plug-in motivates a bias-corrected version, which we do not develop. The scale-versus-shape and width diagnostics of Section 3 are numerical. Finally, the main analysis is for balanced complete randomization; Remark 7 extends it to designs whose imbalance is of order $\sqrt{n}$, but other treatment probabilities and designs remain open.

5 Conclusion

In balanced randomized experiments REG and IREG have the same first-order asymptotic variance, so their relative performance cannot be determined from standard efficiency theory alone. The main contribution of this paper is a closed-form second-order comparison: the difference in the $1/n^2$ terms of their expansion MSEs depends on treatment-effect heterogeneity, covariate shape, residual heteroskedasticity and nonlinear misspecification, can favor either estimator, and translates into an effective sample-size difference that ranges from a few observations to more than a hundred in our skewed scalar designs; in the most heavy-tailed designs the finite-sample gaps are larger still, but the expansion no longer predicts their magnitude.

As a practical companion, our simulations show that confidence intervals from standard software behave differently for the two estimators in finite samples. With no treatment-effect heterogeneity, interacted regression with HC1 standard errors can undercover substantially, in a pattern consistent with high within-arm leverage; HC3 substantially improves its coverage in most designs but not in the most extreme ones; and simple regression with HC1 is close to nominal at the 95% level in the designs examined. Under superpopulation sampling, the variance of centered interacted regression for the population average treatment effect has an additional component, not included in its conventional robust variance, that matters when covariate-explained heterogeneity is large; in our large-heterogeneity design its plug-in estimate tracks the true component and restores calibration, with substantially wider intervals that reflect the added uncertainty, whereas with little or no heterogeneity the plug-in is noisy and miscalibrated across nominal levels.

These inference results are simulation evidence, not theorems. A higher-order theory of robust-variance coverage for regression adjustment, designs with a growing number of covariates, and unbalanced designs beyond the local extension of Remark 7 remain outside the scope of the paper.

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A Proofs

Throughout this appendix, Latin indices i, j, k, l, a, b, c, d attached to $Z, T_3, T_4, \mathcal{E}$ and P refer to *coordinates* of the whitened covariate vector and are summed over in the Einstein convention; the unit index is suppressed. We write $G = \|\gamma\|^2$.

A.1 Second-order expansion

Under complete randomization with $n_1 = m$ treated units, write $U_i^{(w)}$ for the vector collecting $X_i, Y_i(w)$ and the distinct entries of $X_i X_i^\top$ and $X_i Y_i(w)$, and let $\bar{U}_1$ and $\bar{U}_0$ be the averages of $U_i^{(1)}$ over treated units and of $U_i^{(0)}$ over control units, with population means μ_1 and μ_0 .

Lemma 8 (Two-sample representation). *Let $(X_i, Y_i(0), Y_i(1))$ be IID and let the assignment vector W be independent of them, with $\sum_i W_i = m$. Conditionally on W , the treated observations $\{(X_i, Y_i(1)) : W_i = 1\}$ and the control observations $\{(X_i, Y_i(0)) : W_i = 0\}$ are independent IID samples of sizes m and $n - m$ from the laws of $(X, Y(1))$ and $(X, Y(0))$. Both $\hat{\tau}_{\text{REG}}$ and $\hat{\tau}_{\text{IREG}}$ are of the form $h_{j,r}(\bar{U}_1, \bar{U}_0)$ with $r = m/n$, where $h_{j,r}$ is infinitely differentiable in a neighborhood of (μ_1, μ_0) and its derivatives at (μ_1, μ_0) are rational functions of r without poles in $(0, 1)$.*

Proof. Given W , the units remain IID because W is independent of them; the treated and control index sets are fixed and disjoint, so the two blocks are independent IID samples, observed through $Y_i(1)$ and $Y_i(0)$ respectively. Both estimators are rational functions of per-arm means, second moments and cross-moments: $\hat{\tau}_{\text{IREG}}$ is the difference of the arm-wise intercepts after centering at $\bar{X} = r\bar{X}_1 + (1-r)\bar{X}_0$, with arm-wise least-squares slopes $\mathcal{S}_w^{-1}s_w$ built from $\bar{U}_w$; $\hat{\tau}_{\text{REG}}$ uses the pooled within-arm slope, whose Gram matrix is $rS_1 + (1-r)S_0$. The Gram matrices are invertible at (μ_1, μ_0) because $\Sigma \succ 0$, which gives the smoothness and the dependence on r . $\square$

Lemma 9 (Conditional expansion). *Under Assumption 1, let $\text{MSE}_{\text{CRT},r}^{\text{exp}}(\hat{\tau}_j)$ be the second moment of the third-order Taylor polynomial of $h_{j,r}$ at (μ_1, μ_0) . For every $\eta \in (0, \frac{1}{2})$,*

$$\text{MSE}_{\text{CRT},r}^{\text{exp}}(\hat{\tau}_j) = \frac{V_j(r)}{n} + \frac{c_j(r)}{n^2} + O(n^{-3}), \quad j \in \{\text{REG}, \text{IREG}\}, \quad (12)$$

uniformly in $r = m/n \in [\frac{1}{2} - \eta, \frac{1}{2} + \eta]$, with V_j and c_j infinitely differentiable in r .

Proof. The Taylor polynomial is a polynomial of degree three in $(\bar{U}_1 - \mu_1, \bar{U}_0 - \mu_0)$ whose coefficients are smooth in r (Lemma 8); its square has degree six. By Lemma 8 the two blocks are independent IID samples, so each mixed moment of the centered per-arm means is a finite sum of products of joint cumulants of $U^{(1)}$ and $U^{(0)}$ divided by powers of $m = rn$ and $n - m = (1 - r)n$. Moments up to order six are finite because the entries of $U^{(w)}$ are products of at most two coordinates of $(X_i, Y_i(w))$ and Assumption 1 bounds their twelfth moments. Collecting terms gives a finite expansion in powers of $1/n$ whose coefficients are rational in r without poles on $[\frac{1}{2} - \eta, \frac{1}{2} + \eta]$; the terms beyond n^{-2} are $O(n^{-3})$ uniformly on that interval. $\square$

Lemma 10 (Bernoulli and conditional expansions). *Under Assumption 1 with $W_i \stackrel{\text{iid}}{\sim} \text{Bernoulli}(1/2)$ independent of the units, let $\bar{W} = n^{-1} \sum_i W_i$. For every $\eta \in (0, \frac{1}{2})$,*

$$\text{MSE}_{\text{Bern}}^{\text{exp}}(\hat{\tau}_j) = \mathbb{E} \left[\text{MSE}_{\text{CRT},\bar{W}}^{\text{exp}}(\hat{\tau}_j) \mathbf{1}\{|\bar{W} - \frac{1}{2}| \leq \eta\} \right] + O(n^{-5/2}).$$

Proof. Let $\mathcal{P}_{\text{Bern}}$ be the Taylor polynomial of Definition 1 under Bernoulli assignment and $\mathcal{P}_{\bar{W}}$ the conditional polynomial of Lemma 9 evaluated at $r = \bar{W}$. Every coordinate of $\bar{\mathcal{V}} - \mu$ is a polynomial of degree at most two in $(\bar{W} - \frac{1}{2}, \bar{U}_1 - \mu_1, \bar{U}_0 - \mu_0)$ and is linear in the per-arm means; for instance $n^{-1} \sum_i W_i U_i^{(1)} - \frac{1}{2} \mu_1 = (\bar{W} - \frac{1}{2}) \mu_1 + \frac{1}{2} (\bar{U}_1 - \mu_1) + (\bar{W} - \frac{1}{2}) (\bar{U}_1 - \mu_1)$. Hence $\mathbb{E}[(\mathcal{P}_{\text{Bern}} - \tau)^2 \mid W]$ is bounded uniformly in W , and since $\Pr(|\bar{W} - \frac{1}{2}| > \eta) \leq 2e^{-2n\eta^2}$ by Hoeffding's inequality, the event $|\bar{W} - \frac{1}{2}| > \eta$ contributes $O(e^{-2n\eta^2})$. On its complement, write $\hat{\tau}_j = F_j(\bar{W} - \frac{1}{2}, \bar{U}_1 - \mu_1, \bar{U}_0 - \mu_0)$ with F_j smooth. Both $\mathcal{P}_{\text{Bern}}$ and $\mathcal{P}_{\bar{W}}$ agree with the Taylor expansion of F_j at the origin through total degree three, $\mathcal{P}_{\text{Bern}}$ because the map to $\bar{\mathcal{V}}$ is polynomial and $\mathcal{P}_{\bar{W}}$ because it is exact in $\bar{W}$, so $\mathcal{P}_{\text{Bern}} - \mathcal{P}_{\bar{W}}$ consists of terms of total degree at least four, each of which is at most cubic in the per-arm means. Conditionally on $\bar{W}$, the centered per-arm means are $O(n^{-1/2})$ in the L^2 norms that arise, and $\bar{W} - \frac{1}{2} = O(n^{-1/2})$ in every L^k ; hence $\|\mathcal{P}_{\text{Bern}} - \mathcal{P}_{\bar{W}}\|_{L^2} = O(n^{-2})$ and $\|\mathcal{P}_{\text{Bern}} + \mathcal{P}_{\bar{W}} - 2\tau\|_{L^2} = O(n^{-1/2})$, and the Cauchy-Schwarz inequality gives $|\mathbb{E}[(\mathcal{P}_{\text{Bern}} - \tau)^2 - (\mathcal{P}_{\bar{W}} - \tau)^2]| = O(n^{-5/2})$ on that event. $\square$

The proof of Theorem 2 combines these lemmas, a reduction from the CRT to the Bernoulli design, and the Bernoulli calculation.

Theorem 11. *Suppose the Bernoulli design has second-order difference*

$$\text{MSE}_{\text{Bern}}^{\text{exp}}(\hat{\tau}_{\text{IREG}}) - \text{MSE}_{\text{Bern}}^{\text{exp}}(\hat{\tau}_{\text{REG}}) = \frac{D_{\text{Bern}}}{n^2} + O(n^{-5/2}) \quad (13)$$

at $\pi = 1/2$, and let Assumption 1 hold. Then, under complete randomization with $n_1 = \lfloor n/2 \rfloor$ treated units,

$$\text{MSE}_{\text{CRT}}^{\text{exp}}(\hat{\tau}_{\text{IREG}}) - \text{MSE}_{\text{CRT}}^{\text{exp}}(\hat{\tau}_{\text{REG}}) = \frac{D_{\text{Bern}} + 4G}{n^2} + O(n^{-5/2}). \quad (14)$$

Proof. Let $n_1 = \sum_{i=1}^n W_i$ under the Bernoulli design and write $r = n_1/n$. Conditional on $n_1 = m$, the treatment assignment has the same distribution as a CRT with treatment fraction r . Fix $\eta \in (0, \frac{1}{2})$. By Lemma 10, $\text{MSE}_{\text{Bern}}^{\text{exp}}$ is, up to $O(n^{-5/2})$, the average over n_1 of $\text{MSE}_{\text{CRT},r}^{\text{exp}}$ restricted to $|r - \frac{1}{2}| \leq \eta$, and by Lemma 9 the latter admits the expansion (12) uniformly on that set, with V_j and c_j infinitely differentiable. At treatment fraction r the first-order variance difference is, by Proposition 1,

$$V_{\text{IREG}}(r) - V_{\text{REG}}(r) = -\frac{(1-2r)^2}{r(1-r)}G, \quad \frac{(1-2r)^2}{r(1-r)} = 16u^2 + O(u^4), \quad u = r - \frac{1}{2}.$$

Under Bernoulli randomization with $\pi = 1/2$, $\mathbb{E}[u^2] = 1/(4n)$ and $\mathbb{E}[u^4] = O(n^{-2})$, so

$$\frac{1}{n} \mathbb{E}[V_{\text{IREG}}(r) - V_{\text{REG}}(r)] = -\frac{4G}{n^2} + O(n^{-3}).$$

For the second-order coefficients, a second-order Taylor expansion of $c_{\text{IREG}} - c_{\text{REG}}$ around $1/2$ together with $\mathbb{E}[u] = 0$ and $\mathbb{E}[u^2] = O(n^{-1})$ gives $\mathbb{E}[c_{\text{IREG}}(r) - c_{\text{REG}}(r)] = c_{\text{IREG}}(1/2) - c_{\text{REG}}(1/2) + O(n^{-1})$. Averaging (12) over n_1 therefore yields

$$\text{MSE}_{\text{Bern}}^{\text{exp}}(\hat{\tau}_{\text{IREG}}) - \text{MSE}_{\text{Bern}}^{\text{exp}}(\hat{\tau}_{\text{REG}}) = \frac{c_{\text{IREG}}(1/2) - c_{\text{REG}}(1/2) - 4G}{n^2} + O(n^{-5/2}),$$

so $D_{\text{Bern}} = c_{\text{IREG}}(1/2) - c_{\text{REG}}(1/2) - 4G$. Under balanced complete randomization $r = 1/2$ for even n , giving $D_{\text{CRT}} = c_{\text{IREG}}(1/2) - c_{\text{REG}}(1/2) = D_{\text{Bern}} + 4G$. For odd n , $r - 1/2 = O(n^{-1})$, so the induced change in the first-order variance difference is $O(n^{-3})$ and does not affect the n^{-2} coefficient. $\square$

Theorem 12. Let Assumption 1 hold, except that $W_i \stackrel{\text{iid}}{\sim} \text{Bernoulli}(1/2)$ independently of $(X_i, Y_i(0), Y_i(1))$. Define

$$T_{3,ijk} = \mathbb{E}[Z_i Z_j Z_k], \quad T_{4,ijkl} = \mathbb{E}[Z_i Z_j Z_k Z_l], \quad q_i = T_{3,ijk} \delta_{jk}, \quad L_{ij} = T_{3,iab} T_{3,jab}, \quad (15)$$

write $S = q = \mathbb{E}[Z\|Z\|^2]$, and set $\mathcal{E} = \mathbb{E}[ZZ^\top \varepsilon]$ and $P = \mathbb{E}[ZZ^\top (W - \frac{1}{2})\varepsilon]$. For the scalar contractions

$$G = \|\gamma\|^2, \quad F = \gamma_i \gamma_j T_{4,ijkk}, \quad Q = (\gamma^\top q)^2, \quad \mathcal{R} = \gamma^\top L \gamma, \quad (16)$$

$$A_P = \gamma_i T_{3,ijk} P_{jk}, \quad B_P = (\gamma^\top q) \text{tr}(P), \quad C_P = \gamma^\top P q, \quad (17)$$

we have

$$\text{MSE}_{\text{Bern}}^{\text{exp}}(\hat{\tau}_{\text{IREG}}) - \text{MSE}_{\text{Bern}}^{\text{exp}}(\hat{\tau}_{\text{REG}}) = \frac{D_{\text{Bern}}}{n^2} + O(n^{-5/2}), \quad (18)$$

where

$$D_{\text{Bern}} = -(2p+7)G + F - Q - 3\mathcal{R} - 16A_P - 8B_P + 8C_P + 8\text{tr}(\mathcal{E}^2) + 8S^\top \mathbb{E}[Z\varepsilon^2]. \quad (19)$$

Proof. Both REG and IREG contain the regressors 1, W and X . Subtracting any linear combination of these regressors from the outcome changes the fitted coefficient on W by the corresponding amount while leaving the estimation error unchanged, so we may subtract the population main effects and the term τW_i and work with the exact representation $Y_i = W_i X_i^\top \{\beta_{(1)}^* - \beta_{(0)}^*\} + \varepsilon_i$. Whitening as in (3) gives the canonical representation $Y_i = W_i \gamma^\top Z_i + \varepsilon_i$, and we write

$$W_i = \frac{1}{2} + \omega_i, \quad \omega_i \in \{-\frac{1}{2}, \frac{1}{2}\}, \quad \mathbb{E}[\omega_i] = 0, \quad \omega_i^2 = \frac{1}{4}.$$

Let $\mathcal{V}_i$ denote the vector of raw-moment coordinates needed to evaluate the REG or IREG coefficient map, and adopt the notation of Definition 1, so that $\hat{\tau} = g(\bar{\mathcal{V}})$ and $\tilde{\tau}_n$ is given by (8). Let

$$\Sigma_{ij}^\mathcal{V} = \text{Cov}(\mathcal{V}_i, \mathcal{V}_j), \quad \mu_{3,ijk}^\mathcal{V} = \mathbb{E}[(\mathcal{V}_i - \mu_i)(\mathcal{V}_j - \mu_j)(\mathcal{V}_k - \mu_k)]$$

be the covariance and third central-moment tensors of the moment vector. Expanding the square of (8) and using the second- and third-order moments of sample means gives the order- n^{-1} correction to $n \text{MSE}^{\text{exp}}$ as $\mathcal{T}_1 + \mathcal{T}_2 + \mathcal{T}_3$ with

$$\mathcal{T}_1 = a_i H_{jk} \mu_{3,ijk}^\mathcal{V}, \quad (20)$$

$$\mathcal{T}_2 = \frac{1}{4} H_{ij} H_{kl} (\Sigma_{ik}^\mathcal{V} \Sigma_{jl}^\mathcal{V} + \Sigma_{il}^\mathcal{V} \Sigma_{jk}^\mathcal{V} + \Sigma_{ij}^\mathcal{V} \Sigma_{kl}^\mathcal{V}), \quad (21)$$

$$\mathcal{T}_3 = \frac{1}{3} a_i A_{3,jkl} (\Sigma_{ij}^\mathcal{V} \Sigma_{kl}^\mathcal{V} + \Sigma_{ik}^\mathcal{V} \Sigma_{jl}^\mathcal{V} + \Sigma_{il}^\mathcal{V} \Sigma_{jk}^\mathcal{V}), \quad (22)$$

the remaining terms being $O(n^{-3/2})$ in $n \text{MSE}^{\text{exp}}$ under Assumption 1. The difference between the order- n^{-2} MSE coefficients of REG and IREG is obtained by evaluating these three contractions for the two estimators.

The population Gram matrices and their inverses in the whitened coordinates are

$$A_R = \begin{pmatrix} 1 & 0 & 1/2 \\ 0 & I_p & 0 \\ 1/2 & 0 & 1/2 \end{pmatrix}, \quad A_R^{-1} = \begin{pmatrix} 2 & 0 & -2 \\ 0 & I_p & 0 \\ -2 & 0 & 4 \end{pmatrix}, \quad A_I = \begin{pmatrix} 1 & 0 & 1/2 & 0 \\ 0 & I_p & 0 & I_p/2 \\ 1/2 & 0 & 1/2 & 0 \\ 0 & I_p/2 & 0 & I_p/2 \end{pmatrix},$$

$$A_I^{-1} = \begin{pmatrix} 2 & 0 & -2 & 0 \\ 0 & 2I_p & 0 & -2I_p \\ -2 & 0 & 4 & 0 \\ 0 & -2I_p & 0 & 4I_p \end{pmatrix}, \quad \theta_R = (0, \gamma/2, 0), \quad \theta_I = (0, 0, 0, \gamma).$$

Writing $\Omega = A^{-1}$ and $\theta = \Omega b$, differentiation of $A\theta = b$ gives the exact derivative identities

$$D\theta[h] = \Omega(db - dA\theta), \quad D^2\theta[h_1, h_2] = -\Omega dA_1\Omega(db_2 - dA_2\theta) - \Omega dA_2\Omega(db_1 - dA_1\theta),$$

$$D^3\theta[h_1, h_2, h_3] = \sum_{\varsigma \in \mathfrak{S}_3} \Omega dA_{\varsigma_1}\Omega dA_{\varsigma_2}\Omega(db_{\varsigma_3} - dA_{\varsigma_3}\theta).$$

REG's estimator is the treatment coefficient, $g_{\text{REG}}(v) = \theta_W(v)$. IREG's is $g_{\text{IREG}}(v) = \theta_W(v) + \bar{z}(v)^\top \theta_{WZ}(v)$, where $\bar{z}(v)$ is the full-sample covariate mean, a linear coordinate of v that vanishes at μ in whitened coordinates, and $\theta_{WZ} = \gamma$ at μ . The product rule gives

$$a_{\text{IREG}}[h] = D\theta_W[h] + D\bar{z}[h]^\top \gamma,$$

$$H_{\text{IREG}}[h_1, h_2] = D^2\theta_W[h_1, h_2] + D\bar{z}[h_1]^\top D\theta_{WZ}[h_2] + D\bar{z}[h_2]^\top D\theta_{WZ}[h_1],$$

$$A_{3,\text{IREG}}[h_1, h_2, h_3] = D^3\theta_W[h_1, h_2, h_3] + \sum_{\{i,j,k\}=\{1,2,3\}} D\bar{z}[h_i]^\top D^2\theta_{WZ}[h_j, h_k],$$

the sum running over the three choices of i . The covariance and third-moment blocks of $\mathcal{V}_i$ are products of moments of $(Z_i, \omega_i, \varepsilon_i)$ under the canonical representation of Section 2, with W_i marginally Bernoulli(1/2).

Substituting these derivative tensors and the covariance and third-moment blocks of $\mathcal{V}_i$ into (20)–(22), and reducing the resulting indexed sums using $\mathbb{E}[\omega_i] = 0$, $\omega_i^2 = 1/4$, the four population normal equations for ε_i and the symmetries of T_3 and T_4 , gives

$$\Delta\mathcal{T}_1 = 2G + 2F + 8\Xi_\gamma, \quad (23)$$

$$\Delta\mathcal{T}_2 = -3G - F - Q - \mathcal{R} - 8\Xi_\gamma - 8A_P - 8B_P, \quad (24)$$

$$\Delta\mathcal{T}_3 = -(2p+6)G - 2\mathcal{R} - 8A_P + 8C_P + 8\text{tr}(\mathcal{E}^2) + 8S^\top \mathbb{E}[Z\varepsilon^2], \quad (25)$$

where $\Xi_\gamma = \mathbb{E}[(\gamma^\top Z)\|Z\|^2\omega\varepsilon]$ with $\omega = W - \frac{1}{2}$. The algebra in (23)–(25) is carried out symbolically: the reference implementation enumerates the full Cartesian products of the expansion directions, applies the derivative tensors above, and reduces the resulting indexed expressions using only tensor symmetries, dummy-index relabeling and the identities implied by $\mathbb{E}[ZZ^\top] = I_p$ and $\omega_i^2 = 1/4$, with the dimension p kept symbolic. The symbolic remainder for each of the three rows is identically zero; the certificate is `verification/general_p_symbolic_certificate.py` in the replication repository and runs in under a minute.

Summing the three rows, the Ξ_γ terms cancel and (19) follows.

The division of labour is as follows. Derived by hand and displayed above: the expansion terms (20)–(22), the Gram matrices and coefficients at μ , the derivative identities, the product-rule structure of the IREG centering functional, and the reduction rules. Machine-checked: the substitution of these objects into (20)–(22) and the reduction of the resulting indexed sums to the rows (23)–(25), which `verification/general_p_symbolic_certificate.py` carries out for symbolic p with a zero remainder. The resulting constant is checked independently by Monte Carlo in Appendix B. $\square$

Theorems 11 and 12 give

$$D_p = -(2p+3)G + F - Q - 3\mathcal{R} - 16A_P - 8B_P + 8C_P + 8\text{tr}(\mathcal{E}^2) + 8S^\top \mathbb{E}[Z\varepsilon^2], \quad (26)$$

which Lemma 13 rewrites in the matrix form of Theorem 2.

Lemma 13 (Equivalent matrix form). *With K , S , S_γ , E and $\mathcal{E}_W$ as in Theorem 2,*

$$G = \|\gamma\|^2, \quad F = \gamma^\top K \gamma, \quad Q = (\gamma^\top S)^2, \quad \mathcal{R} = \|S_\gamma\|_F^2,$$

$$A_P = \text{tr}(S_\gamma \mathcal{E}_W), \quad B_P = (\gamma^\top S) \text{tr}(\mathcal{E}_W), \quad C_P = \gamma^\top \mathcal{E}_W S.$$

Proof. First, $F = \gamma_i \gamma_j T_{4,ijkl} \delta_{kl} = \gamma^\top \mathbb{E}[ZZ^\top \|Z\|^2] \gamma = \gamma^\top K \gamma$, and $S = \mathbb{E}[Z_i Z_j Z_j]_{i=1}^p = q$ gives $Q = (\gamma^\top q)^2 = (\gamma^\top S)^2$. Next, $(S_\gamma)_{ab} = \mathbb{E}[Z_a Z_b (Z^\top \gamma)] = T_{3,abk} \gamma_k$, so

$$\|S_\gamma\|_F^2 = \sum_{a,b} (S_\gamma)_{ab}^2 = \gamma_k \gamma_\ell T_{3,abk} T_{3,abl} = \gamma_k \gamma_\ell L_{k\ell} = \mathcal{R}.$$

Since S_γ and $\mathcal{E}_W$ are symmetric, $\text{tr}(S_\gamma \mathcal{E}_W) = (S_\gamma)_{jk} (\mathcal{E}_W)_{jk} = \gamma_i T_{3,ijk} P_{jk} = A_P$; finally $B_P = (\gamma^\top q) \text{tr}(P) = (\gamma^\top S) \text{tr}(\mathcal{E}_W)$ and $C_P = \gamma^\top P q = \gamma^\top \mathcal{E}_W S$. $\square$

We close with an exact identity that is used by the variance-reduced verification of Appendix B.

Lemma 14 (Exact difference identity). *Let $\bar{X}_w$ and $S_w = \sum_{i:W_i=w} (X_i - \bar{X}_w)(X_i - \bar{X}_w)^\top$ denote the within-arm means and scatter matrices, let $\hat{\beta}_w = S_w^{-1} \sum_{i:W_i=w} (X_i - \bar{X}_w)(Y_i - \bar{Y}_w)$, and write $d_X = \bar{X}_1 - \bar{X}_0$. Then, whenever S_0 , S_1 and $S_0 + S_1$ are invertible and $n_1 = n_0$,*

$$\hat{\tau}_{\text{REG}} = \bar{Y}_1 - \bar{Y}_0 - d_X^\top (S_1 + S_0)^{-1} (S_1 \hat{\beta}_1 + S_0 \hat{\beta}_0), \quad \hat{\tau}_{\text{IREG}} = \bar{Y}_1 - \bar{Y}_0 - d_X^\top \frac{\hat{\beta}_1 + \hat{\beta}_0}{2},$$

and consequently

$$\hat{\tau}_{\text{IREG}} - \hat{\tau}_{\text{REG}} = \frac{1}{2} d_X^\top (S_1 + S_0)^{-1} (S_1 - S_0) (\hat{\beta}_1 - \hat{\beta}_0) = O_P(n^{-1}).$$

Proof. Both expressions for the estimators are the Frisch–Waugh–Lovell representations of (1): the coefficient on X in the REG regression is the pooled within-arm slope $(S_1 + S_0)^{-1} (S_1 \hat{\beta}_1 + S_0 \hat{\beta}_0)$, and IREG fits the two arms separately, so with $n_1 = n_0$ the overall covariate mean satisfies $\bar{X}_1 - \bar{X} = d_X/2 = -(\bar{X}_0 - \bar{X})$. Subtracting,

$$\hat{\tau}_{\text{IREG}} - \hat{\tau}_{\text{REG}} = d_X^\top \left[(S_1 + S_0)^{-1} (S_1 \hat{\beta}_1 + S_0 \hat{\beta}_0) - \frac{1}{2} (\hat{\beta}_1 + \hat{\beta}_0) \right] = \frac{1}{2} d_X^\top (S_1 + S_0)^{-1} (S_1 - S_0) (\hat{\beta}_1 - \hat{\beta}_0).$$

The stochastic order follows from $d_X = O_P(n^{-1/2})$, $(S_1 + S_0)^{-1} (S_1 - S_0) = O_P(n^{-1/2})$ and $\hat{\beta}_1 - \hat{\beta}_0 = O_P(1)$. $\square$

In whitened coordinates the leading term of this difference is $\frac{1}{4} d_Z^\top (A_1 - A_0) \gamma$, with A_w the arm- w mean of $ZZ^\top - I_p$; this is the control variate used in (27) below.

Lemma 15 (Leading expansion bias). *Under Assumption 1 and balanced complete randomization,*

$$\mathbb{E}[\tilde{\tau}_{n,\text{IREG}} - \tau] = -\frac{4 \text{tr}(\mathcal{E}_W)}{n} + O(n^{-2}), \quad \mathbb{E}[\tilde{\tau}_{n,\text{REG}} - \tau] = -\frac{4 \text{tr}(\mathcal{E}_W) + \gamma^\top S}{n} + O(n^{-2}).$$

Proof. Work in whitened coordinates with $m = n/2$ units per arm, arm slopes $b_1 - b_0 = \gamma$, projection residuals $\varepsilon_i(w)$ and within-arm scatter matrices S_w . By Remark 3, the expansion bias is the mean of the quadratic part of $\hat{\tau} - \tau$, up to $O(n^{-2})$. For IREG, $\bar{Z}_1 - \bar{Z} = -(\bar{Z}_0 - \bar{Z}) = \frac{1}{2}(\bar{Z}_1 - \bar{Z}_0)$, so

$$\hat{\tau}_{\text{IREG}} - \tau = \bar{\varepsilon}_1 - \bar{\varepsilon}_0 + \gamma^\top \bar{Z} - \frac{1}{2} (\bar{Z}_1 - \bar{Z}_0)^\top \{(\hat{b}_1 - b_1) + (\hat{b}_0 - b_0)\},$$

and the linear part of $\hat{b}_w - b_w$ is $m^{-1} \sum_{i:W_i=w} Z_i \varepsilon_i(w)$, which enters the quadratic part of $\hat{\tau}_{\text{IREG}} - \tau$ through its product with $\bar{Z}_1 - \bar{Z}_0$. The arms are independent (Lemma 8), so $\mathbb{E}[(\bar{Z}_1 - \bar{Z}_0)^\top (\hat{b}_1 - b_1)] = m^{-1} \mathbb{E}[\|Z\|^2 \varepsilon(1)] + O(n^{-2})$ and $\mathbb{E}[(\bar{Z}_1 - \bar{Z}_0)^\top (\hat{b}_0 - b_0)] = -m^{-1} \mathbb{E}[\|Z\|^2 \varepsilon(0)] + O(n^{-2})$, which gives $-(2m)^{-1} \text{tr}\{\mathbb{E}[ZZ^\top \varepsilon(1)] - \mathbb{E}[ZZ^\top \varepsilon(0)]\} = -4 \text{tr}(\mathcal{E}_W)/n$. For REG, the pooled slope $\hat{b}$ estimates $\bar{b} = (b_1 + b_0)/2$ and

$$\hat{\tau}_{\text{REG}} - \tau = \bar{\varepsilon}_1 - \bar{\varepsilon}_0 + \frac{1}{2} \gamma^\top (\bar{Z}_1 + \bar{Z}_0) - (\bar{Z}_1 - \bar{Z}_0)^\top (\hat{b} - \bar{b}).$$

The pooled residual of unit i is $\varepsilon_i(W_i) + (W_i - \frac{1}{2}) \gamma^\top Z_i$ up to centering, so the linear part of $\hat{b} - \bar{b}$ is $n^{-1} \sum_i Z_i \{\varepsilon_i(W_i) + (W_i - \frac{1}{2}) \gamma^\top Z_i\}$, which again enters the quadratic part of $\hat{\tau}_{\text{REG}} - \tau$ through its product with $\bar{Z}_1 - \bar{Z}_0$, and $\mathbb{E}[(\bar{Z}_1 - \bar{Z}_0)^\top (\hat{b} - \bar{b})] = n^{-1} \{\text{tr} \mathbb{E}[ZZ^\top \varepsilon(1)] - \text{tr} \mathbb{E}[ZZ^\top \varepsilon(0)] + \gamma^\top S\} + O(n^{-2})$, which gives the second display. $\square$

B Numerical verification of Theorem 2

The symbolic certificate establishes the algebra of Theorem 12 exactly. The role of simulation is therefore not to re-derive D_p but to provide an independent numerical check that the constant governs the sampling behavior of the estimators: that we expanded the right quantity, that the CRT correction of Theorem 11 is right, and that no term was lost in translation. Simulation cannot prove the theorem; we design the checks so that they can fail, and report their power.

Exact targets. The verification uses data-generating processes whose whitened covariates have independent coordinates with known moments and whose oracle residuals take the form $\varepsilon(w) = g_w(Z) + s_w(Z)\eta$, with η independent of Z with mean zero and unit variance, g_w a polynomial orthogonal to $(1, Z)$ and s_w^2 a polynomial. Every ingredient of D_p is then a finite sum of moments, computed in exact rational arithmetic, so the target carries no Monte Carlo error; the code also verifies $\mathbb{E}[g_w] = \mathbb{E}[Zg_w] = 0$ so that a mis-specified design cannot pass silently. These distributions are chosen for analytic tractability and are deliberately different from the designs of Section 3.3. Six are used, so that every block of D_p is active in at least one: Gaussian homoskedastic (F only); centered-exponential covariates with heterogeneity ($F, Q, \mathcal{R}$); centered-exponential covariates with heteroskedasticity and $\gamma = 0$ (the $8S^\top \mathbb{E}[Z\varepsilon^2]$ block alone); centered-exponential covariates with heterogeneity, arm-specific misspecification and arm-specific heteroskedasticity (all nine terms, including A_P , B_P and C_P); Gaussian covariates with arm-specific misspecification and heteroskedasticity ($\text{tr}(\mathcal{E}^2)$ and $\mathcal{E}_W$); and a mixed skewed/Gaussian design.

A variance-reduced estimator of D_p . The natural estimator $n^2\{\widehat{\text{MSE}}_{\text{IREG}} - \widehat{\text{MSE}}_{\text{REG}}\}$ is badly behaved: by Lemma 14 the difference of the estimators is $O_P(n^{-1})$ while $\hat{\tau}_{\text{REG}} - \tau = O_P(n^{-1/2})$, so the per-replication standard deviation of the statistic grows like $\sqrt{n}$ and confidence intervals become *wider* as n grows at fixed replication count. We therefore subtract an exactly-centered control variate. With $d_Z = \bar{Z}_1 - \bar{Z}_0$, A_w the arm- w mean of $ZZ^\top - I_p$, and

$$d_1 = \frac{1}{4} d_Z^\top (A_1 - A_0) \gamma, \quad L = \gamma^\top \bar{Z} + \bar{\varepsilon}_1 - \bar{\varepsilon}_0,$$

the expectation of a product of three centered IID sample means over an arm of size $m = n/2$ equals the corresponding third cumulant divided by m^2 , which gives the exact finite- n identity

$$\mathbb{E}[d_1 L] = \frac{1}{n^2} \mathbb{E}\left[\left(\|Z\|^2 - 1\right) (\gamma^\top Z) (\gamma^\top Z + \varepsilon(1) - \varepsilon(0))\right]. \quad (27)$$

The estimator

$$\hat{D}_p = \frac{1}{B} \sum_{b=1}^B \left[n^2 (e_{\text{IREG},b}^2 - e_{\text{REG},b}^2) - 2n^2 d_{1,b} L_b \right] + 2n^2 \mathbb{E}[d_1 L]$$

is unbiased for the same quantity as the raw statistic, with per-replication variance $O(1)$ rather than $O(n)$. Both are reported by the replication code: in the Gaussian design the raw per-replication standard deviation grows from 151 at $n = 200$ to 293 at $n = 800$, while the control-variate standard deviation is 56 and 55; at $n = 3200$ the control-variate standard error is ten times smaller than the raw one.

Results. Table 7 reports the exact targets and the control-variate estimates at $n \in \{200, 800, 3200\}$, with $B = 400,000$ replications at the two smaller sample sizes and $B = 120,000$ at the largest. Because the control-variate estimator has per-replication standard deviation of order one, it resolves the finite- n remainder: at $n = 200$ several designs lie many standard errors from their targets, up to 22 in the design with all blocks active. These deviations shrink as n grows; in that design the gap to the target falls from 12.1 to 5.7 to 2.6 as n quadruples twice. In that design $D_{\text{Bern}} = D_p - 4G = 41.32$ lies within one standard error of the estimate at $n = 3,200$, so this design alone does not separate D_p from D_{Bern} at that sample size. Its successive increments, 6.4 and 3.1, roughly halve as n quadruples, which extrapolates to about 45, close to D_p ; the ablation in Table 8 separates the two decisively in other designs. At $n = 3200$ every estimate is within 2.6 standard errors of its target. The two largest deviations are -2.6 in the all-blocks design, whose trajectory is still approaching the target from below, and $+2.3$ in the $\gamma = 0$ heteroskedastic design, which lies below its target at $n = 200$ and $n = 800$ and above it at $n = 3200$; with six designs, a deviation of this size is unremarkable under Monte Carlo error. The deviations are consistent with a remainder that vanishes relative to n^{-2} in the untrimmed MSE difference, but with three sample sizes we do not attempt to estimate its rate.

Table 7: Verification of Theorem 2. Targets are exact; estimates are the control-variate estimator with Monte Carlo standard errors in parentheses. $B = 400,000$ for $n \in \{200, 800\}$ and $B = 120,000$ for $n = 3200$.

DGP (active blocks)	D_p (exact)	$n = 200$	$n = 800$	$n = 3200$
Gaussian, homoskedastic, $p = 3$ (F)	-5.00	-4.70 (0.09)	-4.99 (0.09)	-5.17 (0.16)
Exponential covariates, homoskedastic (F, Q, R)	-20.25	-17.17 (0.21)	-19.60 (0.22)	-19.75 (0.39)
Exponential, heteroskedastic, $\gamma = 0$	32.00	29.82 (0.28)	31.73 (0.29)	33.25 (0.54)
Exponential, HTE + arm-specific misspec. (all)	44.52	32.40 (0.56)	38.82 (0.55)	41.97 (0.98)
Gaussian, arm-specific misspec. ($\text{tr } E^2, E_W$)	-2.32	-2.36 (0.09)	-2.34 (0.09)	-2.45 (0.16)
Mixed skewed/Gaussian, heteroskedastic	28.00	25.94 (0.27)	27.85 (0.28)	28.27 (0.53)

Power: what the verification would reject. A check that cannot fail is not evidence. Table 8 reports z -statistics of the $n = 3200$ estimates against deliberately perturbed versions of the formula: the Bernoulli constant $D_{\text{Bern}} = D_p - 4G$ (that is, omitting Theorem 11), the Gaussian special case $-(p+1)G$, and D_p with individual blocks deleted. The constant of Theorem 2 has $|z| \leq 2.6$ in every design, while each alternative has $|z| > 3$ in at least one design, in several cases by tens of standard errors. The check has limited power against alternatives that differ from D_p by less than the residual finite- n bias: in the all-blocks design, for example, $D_{\text{Bern}} = D_p - 3.2$ is not separated from D_p at $n = 3200$, which is why several designs are

needed. A conventional Monte Carlo design without the control variate would have standard errors of 2–12 on these quantities and could not separate most of the alternatives.

Table 8: Ablation: $z = (\hat{D}_p - D^{\text{alt}})/\widehat{SE}$ at $n = 3200$ for each design and each candidate constant. Values with $|z| > 3$ indicate rejection of that candidate.

DGP	D_p	$D_p - 4G$	$-(p+1)G$	drop $-3\mathcal{R}$	drop $-Q$	drop resid.	drop $\mathcal{E}_W$
Gaussian, homoskedastic, $p = 3$ (F)	−1.0	+29.8	−1.0	−1.0	−1.0	−1.0	−1.0
Exponential covariates, homoskedastic (F, Q, R)	+1.3	+14.2	−41.3	−37.4	−21.9	+1.3	+1.3
Exponential, heteroskedastic, $\gamma = 0$	+2.3	+2.3	+61.1	+2.3	+2.3	+61.1	+2.3
Exponential, HTE + arm-specific misspec. (all)	−2.6	+0.7	+45.1	−12.4	−3.2	+65.8	−17.7
Gaussian, arm-specific misspec. (tr E^2, E_W)	−0.8	+23.5	+3.4	−0.8	−0.8	+3.4	−0.8
Mixed skewed/Gaussian, heteroskedastic	+0.5	+4.2	+55.7	−5.1	−1.4	+60.3	+0.5

The CRT correction and the bias decomposition. Theorem 11 is checked separately by estimating the n^{-2} constant under both the balanced CRT and Bernoulli(1/2) assignment on the same designs (`verification/crt_correction_certificate.py`): the estimated gap between the two designs is 4.3 ± 0.6 at $n = 200$ and 6.3 ± 1.6 at $n = 800$ in the Gaussian design, and 3.0 ± 1.1 and 8.6 ± 3.0 in the exponential-covariate design, against the predicted $4G = 5$ in both, with all four $|z| < 2$. This check uses the raw statistic, because the control variate is derived for the two-independent-samples representation of the CRT, so its power is modest; the ablation in Table 8, which rejects D_{Bern} by up to thirty standard errors, is the sharper test. Remark 3 is checked by estimating the biases of both estimators directly (`simulations/run_bias.py`): with 10^6 replications, at $n = 800$ all six estimated biases (REG and IREG in three designs) are within 0.8 Monte Carlo standard errors of the leading-order predictions; at $n = 200$ the all-blocks design deviates by about 11% of the predicted REG bias ($z = 9.3$), a gap that has disappeared at $n = 800$ ($z = 0.4$), consistent with the $O(n^{-2})$ remainder of Lemma 15.

C Range of validity of the second-order approximation

Theorem 2 describes the limit $n \rightarrow \infty$; separately from whether D_p is correct, one should ask when $1 + \Delta/n$ is a usable approximation to the finite-sample MSE ratio. Because Δ is driven by third and fourth moments of the whitened covariates, and those same moments control the size of the neglected terms, the two questions are linked. Table 9 makes the trade-off explicit in the scalar lognormal family of the introduction, with $\sigma(X) = 1 + X$ and treatment-effect slope 0.4 held fixed and only the log-scale s of $X \sim \text{LogNormal}(0, s^2)$ varying.

Within this family, for $s = 0.8$ (skewness 3.7) the simulated excess MSE is 75% of the predicted excess Δ/n at $n = 500$ and 85% at $n = 2,000$ (for $s = 0.6$, 91% and 94%); the prediction progressively overstates the gap as the tail becomes heavier; at $s = 1.5$ the predicted ratio exceeds the simulated one by a factor of four at $n = 500$. We do not claim that this boundary transfers to other families of distributions. In the heavy-tailed variant of the introductory example ($s = 1.8$, $\sigma(X) = 1 + 1.5X$, slope 0.3), $\Delta \approx 3.9 \times 10^4$ and $1 + \Delta/n$ exceeds the simulated ratio by a factor of more than forty at $n = 500$: the ordering is right and REG’s advantage is real (the simulated MSE ratio is 1.68 at $n = 500$, 1.34 at $n = 2,000$ and 1.19 at $n = 8,000$, against predictions of 78.5, 20.4 and 5.8), but the deficiency is no longer a useful numerical summary. That regime is also where HC1 intervals fail for IREG: in this design ($R_\tau^2 \approx 0.009$), IREG+HC1 covers τ in 85.6% of replications, IREG+HC2 in 88.9% and IREG+HC3 in 94.0%, against 94.4% for REG+HC1 and 96.7% for REG+HC3.

Table 9: Simulated MSE ratio $\widehat{\text{MSE}}_{\text{IREG}}/\widehat{\text{MSE}}_{\text{REG}}$ versus the second-order prediction $1 + \Delta/n$, for $X \sim \text{LogNormal}(0, s^2)$, $\sigma(X) = 1 + X$, treatment-effect slope 0.4. Monte Carlo standard errors of the simulated ratios are given in parentheses in units of the last reported digit. This family has $\sigma(X) = 1 + X$; the introductory example ($\sigma(X) = 1 + 2X$) and its heavy-tailed variant ($\sigma(X) = 1 + 1.5X$, slope 0.3) are separate designs.

s	skew Z	Δ	$n = 500$		$n = 2000$		$n = 8000$	
			sim.	pred.	sim.	pred.	sim.	pred.
0.6	2.3	4	1.0072 (1)	1.0079	1.0019 (1)	1.0020	1.0004 (1)	1.0005
0.8	3.7	12	1.0184 (3)	1.0243	1.0052 (2)	1.0061	1.0014 (1)	1.0015
1.0	6.2	41	1.0429 (7)	1.0827	1.0137 (4)	1.0207	1.0043 (2)	1.0052
1.2	11.2	166	1.0923 (15)	1.3323	1.0347 (11)	1.0831	1.0127 (8)	1.0208
1.5	33.5	2025	1.2517 (51)	5.0503	1.1163 (44)	2.0126	1.0534 (36)	1.2531

Tail sensitivity. In the heaviest designs the deficiency is dominated by covariate values far in the tail. Table 10 recomputes $1 + \Delta/n$ at $n = 500$ after truncating the lognormal law at a high quantile, for which the moments entering Δ have closed forms. Truncation at the $1 - 10^{-4}$ quantile reduces Δ by one to three orders of magnitude, so Δ is dominated by far-tail moments. The simulations draw between 10^7 and 5×10^7 covariate values and so reach roughly the $1 - 10^{-7}$ quantile, yet even the law truncated at the $1 - 10^{-8}$ quantile gives predictions several times the simulated ratio. In these rows the expansion is inaccurate already at leading order: in the $\gamma = 0$ designs the simulated standard deviation of $\hat{\tau}_{\text{REG}}$ matches $\sqrt{V^*/n}$ at $s = 0.6$ (0.21 against 0.21) but is 0.72 against 0.88 at $s = 1.5$ and 2.46 against 4.90 at $s = 2$ (Table 4), so their finite-sample behavior lies outside the perturbative regime of the expansion. The discrepancy is therefore primarily a failure of the n^{-2} approximation to predict the magnitude of the gap in these designs; Monte Carlo error in the heaviest rows, whose simulated MSEs are dominated by rare draws, is a secondary source of uncertainty. The sign of the comparison agrees in every design.

Table 10: Second-order prediction $1 + \Delta/n$ at $n = 500$ for the full lognormal law and for its truncation at the q quantile, against the simulated MSE ratio (Monte Carlo standard error in parentheses, in units of the last digit).

Design	Simulated ratio	$1 + \Delta/n$			
		full law	$q = 1 - 10^{-4}$	$q = 1 - 10^{-6}$	$q = 1 - 10^{-8}$
Heavy-tailed example ($s = 1.8$)	1.68 (3)	78.5	1.90	7.94	26.6
Table 9, $s = 1.5$	1.25 (1)	5.05	1.36	2.44	3.81
Section 3.3, $\gamma = 0$, $s = 2$	2.00 (6)	672.1	2.45	17.3	97.4

D Implementation and validation of the inference simulations

The simulations of Section 3 are implemented in R (`r/gamma0_sim.R`). For each replication, the design matrices are built exactly as `lm_robust(y ~ w + x)` and `lm_lin(y ~ w, covariates = ~ x)` build them, and the fits are computed by `estimatr`'s own fitting routine, the function those two interfaces call; intervals use the degrees of freedom `estimatr` reports. The main

runs use `estimatr` 1.0.2; the rare-value design is repeated with `estimatr` 2.0.0. The validation script `r/validate_estimatr.R` and the Python cross-check `simulations/crosscheck_r.py` establish the following, under both versions.

- On 100 simulated datasets from four designs, the formula interfaces `lm_robust` and `lm_lin` and the simulation engine give identical estimates, HC1–HC3 standard errors and degrees of freedom (largest relative difference 0), and a base-R matrix implementation of OLS, leverage and HC1–HC3 agrees to 3×10^{-14} .
- On 20 exported datasets, independent Python implementations (NumPy and `statsmodels`) reproduce the estimates, standard errors, degrees of freedom, confidence limits and leverages to 4×10^{-14} .
- With $p = 1$, the estimates agree with the closed-form within-arm expressions of Lemma 14 to 9×10^{-16} .
- When $\gamma = 0$, every unit’s generated treatment effect equals τ up to rounding, so the sample and population ATEs coincide.
- In a rare-value dataset in which the treated arm contains a single rare unit, that unit has $1 - H_{ii} = 1 \times 10^{-16}$ and residual of absolute value 1×10^{-14} . Setting its HC3 contribution to zero gives a standard error of 0.132; `estimatr` 2.0.0 returns 0.132 with a warning, and `estimatr` 1.0.2 returns 0.195, determined by rounding error in the 0/0 ratio.

Leverage-one handling. `estimatr` 1.0.2, used for all main results, evaluates the 0/0 HC2/HC3 contribution of a leverage-one observation in floating point: HC2 is frequently undefined, because the computed $1 - H_{ii}$ can be slightly negative (no interval in 53.6% of the leverage-one replications of the rare-value design with $\sigma = 1 + |X_{i1}|$), and HC3 receives a rounding-determined contribution, so its 94.9% coverage there does not reflect a valid variance estimate. The development version of `estimatr` on GitHub (DESCRIPTION version 2.0.0, commit `e70f3ee`; we do not claim that it corresponds to a CRAN release) drops the contribution of an observation whose computed leverage is at or above one and warns, but one computed just below one still contributes a rounding-determined term; under it IREG+HC3 covers 90.5% and IREG+HC2 85.8% of those replications.

E The superpopulation correction: a calibration diagnostic

This appendix separates the two quantities discussed in Section 3.2: the true first-order component G/n and its plug-in estimate $\hat{\delta}^\top \hat{\Sigma}_X \hat{\delta}/n$. On each simulated dataset we form five intervals for the population ATE from the same IREG estimate and the same degrees of freedom: HC1; HC1 plus the plug-in, as in (11); HC1 plus the true G/n (an oracle, infeasible in practice); HC3; and HC3 plus the plug-in. Only the t critical value changes with the nominal level. When $G = 0$ the oracle interval coincides with the uncorrected one by construction.

Section 3.2 also uses the Gaussian designs of the correction study: $X_i \sim N(0, I_p)$, $Y_i(0) = 0.9 \sum_j X_{ij} + u_i(0)$ and $Y_i(1) = Y_i(0) - u_i(0) + u_i(1) + 1 + 0.3X_{i1}$ with $N(0, 1)$ residuals, in five designs: residuals shared across arms at $(n, p) = (500, 5)$, $(1000, 2)$, $(1000, 5)$ and $(1000, 10)$, and independent across arms at $(n, p) = (500, 5)$. The calibration diagnostic uses three designs, all with $n = 500$. *Large* R_τ^2 : $p = 5$, $X_i \sim N(0, \Sigma)$ with $\Sigma_{jk} = 0.5^{|j-k|}$, $Y_i(0) = 0.9 \sum_j X_{ij} + u_i$ and $Y_i(1) = Y_i(0) + 1 + 3X_{i1}$ with $u_i \sim N(0, 1)$, so that $G = 9$ and $R_\tau^2 = 9/13$; 20,000 replications.

Small R_τ^2 : the heavy-tailed variant of the introductory example ($X \sim \text{LogNormal}(0, 1.8^2)$, $\sigma(X) = 1 + 1.5X$, treatment-effect slope 0.3); 100,000 replications. $\gamma = 0$: the design of Table 1 with $s = 2$; 20,000 replications. The first design was simulated with the NumPy implementation of Appendix D, the second with the implementation used for the introductory example, and the third with `estimatr` 1.0.2. Monte Carlo standard errors are at most 0.4 percentage points.

Table 11 summarizes the 95% intervals. Its last column is the median over replications of the ratio of the width of the HC1-plus-plug-in interval to that of the HC1 interval; it measures how much the feasible correction lengthens a typical interval. Table 12 and Figure 1 report coverage at nominal levels from 80% to 99%.

Table 11: Coverage (%) of nominal 95% intervals for the population ATE, and the median ratio of interval widths (HC1 plus plug-in over HC1).

Design	R_τ^2	HC1	HC1 + plug-in	HC1 + G/n (oracle)	Median width ratio
$\gamma = 0$, skewed ($s = 2$)	0	83.0	95.1	83.0	1.07
Small R_τ^2 , heavy-tailed example	0.009	85.6	94.4	86.5	1.06
Large R_τ^2	0.692	72.7	95.2	95.1	1.80

Table 12: Coverage (%) of intervals for the population ATE at several nominal levels. “Plug-in” adds $\hat{\delta}^\top \hat{\Sigma}_X \hat{\delta} / n$ to the robust variance; “ G/n (oracle)” adds the true first-order component.

Method	80%	90%	95%	98%	99%
<i>$\gamma = 0$, skewed ($s = 2$): $R_\tau^2 = 0$ exactly</i>					
HC1	64.4	75.5	83.0	89.1	91.9
HC1 + plug-in	74.6	88.3	95.1	98.6	99.5
HC1 + G/n (oracle)	64.4	75.5	83.0	89.1	91.9
HC3	79.5	89.2	93.9	96.9	98.0
HC3 + plug-in	85.7	94.6	98.1	99.5	99.8
REG, HC1	76.3	88.8	95.3	98.6	99.5
<i>Small R_τ^2, heavy-tailed example: $R_\tau^2 = 0.009$</i>					
HC1	67.2	78.5	85.6	91.2	93.7
HC1 + plug-in	75.3	88.1	94.4	97.9	99.1
HC1 + G/n (oracle)	68.2	79.5	86.5	91.8	94.2
HC3	79.8	89.3	94.0	97.0	98.2
HC3 + plug-in	84.9	93.6	97.3	99.1	99.6
REG, HC1	76.2	88.3	94.4	97.8	99.0
<i>Large R_τ^2: $R_\tau^2 = 0.692$</i>					
HC1	52.1	63.9	72.7	80.7	85.3
HC1 + plug-in	80.2	90.0	95.2	98.1	99.1
HC1 + G/n (oracle)	80.2	90.0	95.1	98.2	99.2
HC3	52.7	64.6	73.3	81.3	85.7
HC3 + plug-in	80.4	90.1	95.3	98.2	99.1
REG, HC1	80.2	90.2	95.2	98.2	99.1

Tables 11 and 12 and Figure 1 show three regimes. With large R_τ^2 the plug-in averages 1.00 times G/n , and the feasible and oracle intervals are close to nominal at every level (for example 80.2% and 80.2% at 80%, 99.1% and 99.2% at 99%), while HC1 alone undercovers at every level. The median width ratio of 1.80 is what adding a genuinely large variance component

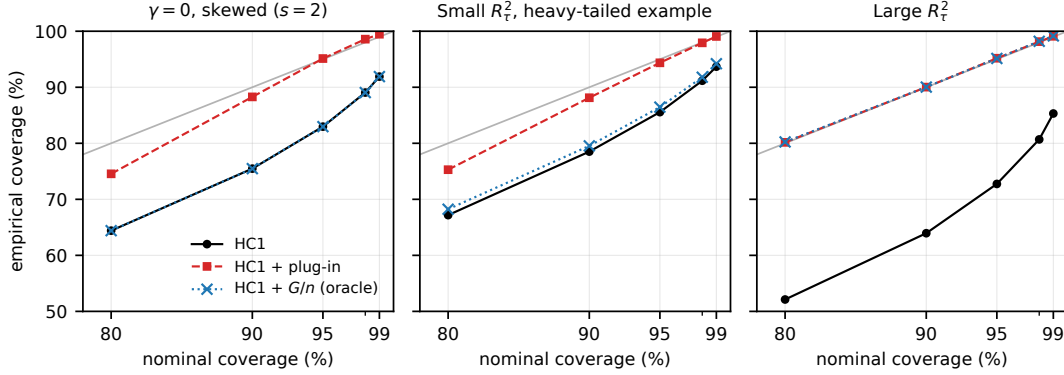


Figure 1: Empirical against nominal coverage for IREG with HC1, HC1 plus the plug-in correction, and HC1 plus the true G/n , in the three designs of Table 12. The grey line is the diagonal. With $\gamma = 0$ (left) the oracle coincides with HC1.

requires. In the small- R_τ^2 example the oracle moves HC1 coverage by about a point, whereas the plug-in, whose average is 31 times G/n , moves it by five to ten points depending on the level; with $\gamma = 0$ the oracle changes nothing and the plug-in again moves coverage substantially. In these two designs the median width ratio is only 1.06 and 1.07: a modest typical lengthening produces large changes in coverage because the HC1 interval itself is badly miscalibrated and the lengthening is concentrated where HC1 fails. With $\gamma = 0$ the median width ratio is 1.48 in replications where the HC1 interval misses and 1.05 where it covers, whereas with large R_τ^2 the lengthening is nearly uniform (1.80 and 1.80), as expected when a genuine variance component is added. Width alone therefore does not establish that an interval is valid or invalid. Across nominal levels the plug-in interval is miscalibrated in these two designs: it undercovers at 80% (75.3% and 74.6%) and slightly overcovers at 99% (99.1% and 99.5%), so its agreement with nominal at 95% is a crossing point rather than evidence of a calibrated procedure; REG+HC1 shows a milder crossing in the $\gamma = 0$ design (Table 12). Here HC3 by itself is within about a point of nominal at every level, and combined with HC3 the plug-in overcovers at every level (85.7% at 80% with $\gamma = 0$). Because $G = 0$ in the last design, all of its effect there comes from the upward bias of the plug-in described in Section 3.2, not from the estimand-level component. Results at 95% for the 20 designs of this paper and 19 supplementary designs are in `results/correction_summary.csv`.